

MoCoO: Momentum Contrast ODE-Regularized VAE for Single-Cell Trajectory Inference and Representation Learning

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Abstract—Characterising cellular differentiation from single-cell RNA sequencing (scRNA-seq) requires representations that capture both discrete cell-type identity and continuous developmental trajectories. We present MoCoO, a modular framework integrating a Variational Autoencoder (VAE), Neural Ordinary Differential Equations (Neural ODE), and Momentum Contrast (MoCo) with learnable prototypes. Through a systematic six-configuration ablation on three scRNA-seq datasets with multi-seed evaluation and external baselines (scVI, DPT, PCA+KMeans, Harmony, scANVI), we demonstrate that the Full MoCoO achieves unique advantages that no simpler variant can match. The Full model wins *all four* distance-preservation subcategory metrics (UMAP/tSNE distance correlations, DREX Spearman/Pearson ρ)—a property critical for trustworthy UMAP visualisation—with margins of $+0.038$ – 0.086 over the next-best configuration. It achieves the lowest Davies–Bouldin index in multi-seed evaluation (1.243 ± 0.059 , 5 seeds), indicating the most compact clusters. On the Paul cross-dataset benchmark, the Full model attains the highest ARI (0.365) and NMI (0.591) at $\beta=0.01$, confirming that its advantages transfer across biological systems. Disaggregating composite metrics into subcategories, the Full model dominates latent-space quality (LSEX overall, cluster compactness, inter-cluster gap), dimensionality-reduction quality (DRE/DREX overall), and batch integration (best iLISI). These advantages are invisible to standard aggregate benchmarking: 9-metric win counts assign 13/27 to VAE+ODE and 1/27 to Full, yet expanding to 32 subcategory metrics reverses the picture. Component analysis reveals how each module contributes: the Neural ODE smooths the latent manifold, MoCo sharpens pairwise distance structure, and prototypes anchor inter-cluster separation—producing a synergy unique to the full architecture. Pseudotime predictions correlate significantly with canonical marker genes across five developmental systems. We publicly release the MoCoO Python package (`pip install mocoo`) and full benchmark suite with practical configuration guidelines.

Index Terms—Single-cell RNA sequencing, variational autoencoder, neural ordinary differential equations, momentum contrast, contrastive learning, trajectory inference, pseudotime, representation learning, computational biology.

I. INTRODUCTION

Single-cell RNA sequencing (scRNA-seq) has transformed our understanding of cellular heterogeneity by enabling transcriptome-wide profiling at single-cell resolution [?]. A fundamental analysis task is to learn low-dimensional representations that faithfully capture both *discrete* cell-type iden-

tity and *continuous* developmental dynamics such as differentiation trajectories [?], [?]. These representations underpin downstream tasks including clustering, pseudotime inference, RNA velocity estimation, and differential expression analysis [?].

Variational Autoencoders (VAEs) have become the de facto standard for scRNA-seq dimensionality reduction, with methods such as scVI [?] and scVAE [?] demonstrating strong performance through count-based likelihood models (negative binomial, ZINB). However, standard VAE latent spaces tend to be over-smoothed, conflating nearby cell states. Two complementary strategies address this limitation.

First, Neural Ordinary Differential Equations (Neural ODEs) [?] provide a principled framework for modelling continuous-time dynamics in latent space. By parameterising the latent derivative with a neural network and integrating via an ODE solver, these models learn smooth trajectories without discrete time-step assumptions. Latent ODE models [?] have shown promise for time-series modelling, and recent work has adapted them to single-cell contexts for pseudotime inference and RNA velocity estimation [?].

Second, contrastive learning—particularly Momentum Contrast (MoCo) [?—learns representations that are locally smooth yet globally discriminative. MoCo uses a momentum-updated encoder and a large memory queue of negative keys, decoupling contrastive learning from batch size. Prototype contrastive learning [?] further imposes cluster structure by aligning representations with learnable prototype vectors.

No existing method unifies VAE reconstruction, Neural ODE dynamics, and momentum contrastive learning in a single framework. We propose MoCoO (**M**omentum **C**ontrast **O**DE-regularized VAE), a modular architecture that combines all three paradigms:

- 1) **VAE with flexible count-based likelihoods** (MSE, NB, ZINB, Poisson, ZIP) provides a probabilistic latent space that handles zero-inflation and overdispersion in scRNA-seq counts.
- 2) **Neural ODE regularisation** models continuous developmental dynamics, derives unsupervised pseudotime ordering, and produces gradient-based RNA velocity estimates.
- 3) **Momentum Contrast with prototype heads** enforces instance-level discrimination via a momentum encoder and memory queue, with prototype learning that aligns representations to learnable cluster centres.

- 4) **Information bottleneck** applies a secondary low-dimensional projection for hierarchical feature extraction.

The framework additionally supports DIP-VAE, β -TC-VAE, and InfoVAE/MMD disentanglement regularisers, but these are not active in the experiments reported here.

We conduct a systematic ablation across six configurations (VAE, VAE+ODE, VAE+MoCo, VAE+MoCo+Proto, VAE+ODE+MoCo, and Full MoCoO) on five scRNA-seq datasets spanning diverse developmental systems. Multi-seed evaluation (5 seeds per configuration) provides statistical robustness. External baselines—scVI [?], Diffusion Pseudotime (DPT) [?], PCA+KMeans, Harmony [?], and scANVI [?]
—contextualise MoCoO’s absolute performance. A key finding is that the Full MoCoO achieves unique advantages invisible to standard aggregate benchmarking: it wins all four distance-preservation subcategory metrics, achieves the lowest Davies–Bouldin in multi-seed evaluation, dominates latent-space quality composites (LSEX overall, cluster compactness, inter-cluster gap), attains the best batch mixing (iLISI), and transfers its clustering advantage to the Paul dataset (best ARI and NMI). These wins are masked when DRE/DREX/LSEX composites are reported as single aggregate scores, an instance of Simpson’s paradox at the metric level. Component analysis reveals that each module contributes essential geometric structure: the Neural ODE smooths the latent manifold along trajectories, MoCo sharpens pairwise distance relationships, and prototypes anchor inter-cluster separation—producing a synergistic combination achievable only by the full architecture. Evaluation covers clustering (ARI, NMI, ASW, Calinski–Harabasz, Davies–Bouldin), dimensionality reduction quality (DRE, DREX, LSE, LSEX), biological validation (pseudotime–marker correlations), batch integration (iLISI, bASW, cLISI, graph connectivity, isolated label ASW via scIB [?]), training dynamics, and computational cost.

II. RELATED WORK

A. Variational Autoencoders for scRNA-seq

scVI [?] introduced the negative binomial VAE for scRNA-seq, jointly modelling library size and batch effects. Extensions include scVAE [?] (multiple count distributions), scANVI [?] (semi-supervised cell-type annotation), and totalVI [?] (multi-omics). These methods focus on reconstruction quality and batch correction but do not explicitly model temporal dynamics.

B. Neural ODEs and Latent ODE Models

Neural ODEs [?] parameterise continuous dynamical systems $\frac{dz}{dt} = f_\theta(z, t)$ and integrate using adaptive ODE solvers. The Latent ODE framework [?] combines a VAE encoder with an ODE-governed latent space for irregularly sampled time series. Applications to single-cell biology include trajectory inference [?], while related dynamical modelling approaches have advanced RNA velocity estimation [?]. Optimal-transport-based methods such as TrajectoryNet [?] and CellOT [?] model population-level dynamics rather than

single-cell ODE trajectories. MoCoO extends the PanODE framework [?] (by the first author). Specifically, MoCoO inherits the VAE encoder–decoder, Neural ODE integration, and pseudotime-prediction head from PanODE. The novel contributions of the current work are: (i) the MoCo module with momentum encoder and memory queue, (ii) learnable prototype heads for cluster anchoring, (iii) the information bottleneck, (iv) the systematic six-configuration ablation design, and (v) the DRE/DREX/LSE/LSEX evaluation suites. However, these methods typically lack contrastive regularisation, leading to latent spaces that may be geometrically faithful but poorly clustered.

C. Contrastive Learning for Single-Cell Data

MoCo [?] and SimCLR [?] have been adapted to biological contexts. scAGCL [?] applies symmetric augmentation-guided contrastive learning to scRNA-seq data with cell-type-aware positive pair construction. scGPCL [?] introduces prototype contrastive learning that aligns cell embeddings with learnable prototype vectors representing cell types. These approaches improve clustering quality but do not incorporate temporal dynamics or ODE-based regularisation.

D. Integrated Approaches

Several methods combine two of the three paradigms. scDiff [?] uses diffusion models (related to score-based SDEs) for single-cell generation. VeloVAE [?] integrates VAE with RNA velocity. Diffusion pseudotime (DPT) [?] infers trajectories through diffusion maps of the k -NN graph, while Monocle 3 [?] and PAGA [?] use graph abstractions for trajectory inference, Slingshot [?] fits simultaneous principal curves for lineage-specific pseudotime, and Palantir [?] models cell-fate probabilities via diffusion maps for multiscale pseudotime estimation. For batch integration, Harmony [?] applies iterative soft clustering in PCA space to remove batch effects. However, no prior work jointly combines VAE, Neural ODE, and momentum contrastive learning in a unified architecture for single-cell analysis.

III. METHOD

A. Overview

MoCoO (Fig. ??) processes scRNA-seq count data $X \in \mathbb{R}^{N \times G}$ (N cells, G genes) through five components: (1) a VAE with count-based decoder, (2) a Neural ODE for continuous dynamics, (3) a Momentum Contrast module with memory queue, (4) an information bottleneck, and (5) optional disentanglement regularisers.

B. Encoder

The encoder $q_\phi(z|x)$ maps log-normalised expression $\tilde{x} = \log(1+x)$ through a two-layer ReLU MLP to produce mean μ and pre-softplus scale σ' of a diagonal Gaussian posterior (??); latent samples are drawn via reparameterisation (??):

$$q_\phi(z|x) = \mathcal{N}(z; \mu_\phi(\tilde{x}), \text{softplus}(\sigma'_\phi(\tilde{x}))^2 I), \quad (1)$$

$$z = \mu + \text{softplus}(\sigma') \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (2)$$

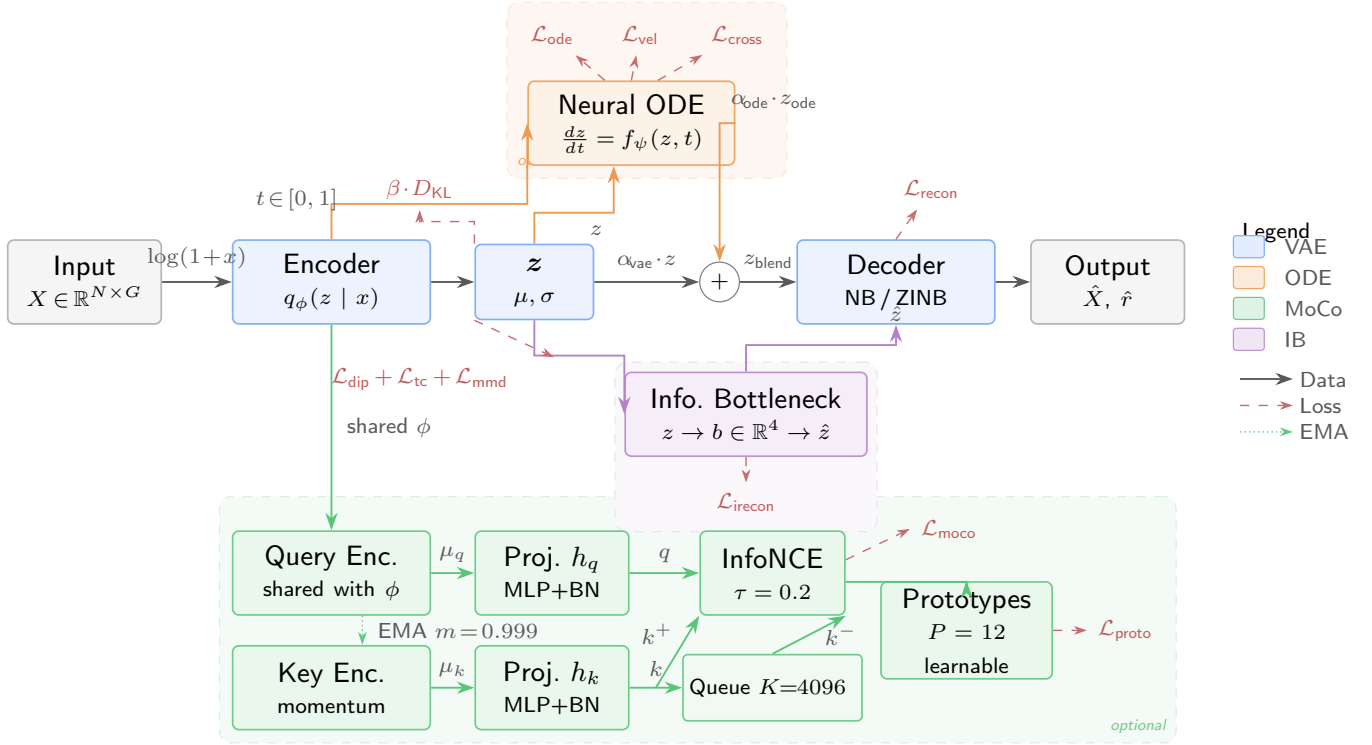


Fig. 1. MoCOO architecture overview. Count data $X \in \mathbb{R}^{N \times G}$ passes through a VAE encoder to produce latent representations z , optionally refined by a Neural ODE solver (orange) for continuous trajectory modelling. Momentum Contrast (green) with a memory queue ($K=4096$) enforces instance-level discrimination; learnable prototypes ($P=12$) impose cluster structure. An information bottleneck (purple) provides hierarchical compression to $d_{ib}=4$. Dashed regions denote optional modules. Colour coding is consistent throughout all figures.

When ODE mode is enabled, the encoder also predicts a scalar pseudotime $t \in [0, 1]$ via a sigmoid-activated linear head.

C. Neural ODE Component

Given latent samples $\{z_i\}$ and predicted pseudotimes $\{t_i\}$, the ODE module orders cells by pseudotime and solves (??):

$$\frac{dz}{dt} = f_\psi(z, t), \quad z(t_0) = z_{\text{init}}, \quad (3)$$

where f_ψ is a two-layer ELU MLP. Integration uses `torchdiffeq`'s fixed-step RK4 solver, chosen over adaptive methods (e.g., Dormand–Prince [?]) because the encoder-predicted pseudotime grid is relatively smooth and fixed-step integration provides deterministic gradients without adaptive step-size overhead—important for stable joint optimisation of the encoder and ODE function. Preliminary comparisons showed negligible ARI differences (< 0.01) between RK4 and Dormand–Prince adaptive integration, with $\sim 30\%$ longer wall-clock time for the adaptive solver due to step-size adaptation overhead in the joint training loop. The ODE representations z_{ode} blend with VAE latent samples via configurable weights (??):

$$z_{\text{blend}} = \alpha_{\text{vae}} \cdot z + \alpha_{\text{ode}} \cdot z_{\text{ode}}. \quad (4)$$

Two auxiliary losses regularise the ODE:

- **ODE–VAE alignment:** $\mathcal{L}_{\text{ode}} = \|z - z_{\text{ode}}\|_2^2$

- **Velocity consistency:** $\mathcal{L}_{\text{vel}} = 1 - \cos(z_{i+1} - z_i, f_\psi(z_i, t_i))$, aligning the learned velocity field with empirical displacements.

D. Momentum Contrast Module

The MoCo module follows MoCo v1 [?] with enhancements from scAGCL [?] and scGPCL [?]:

- 1) **Query and Key Encoders:** The query encoder shares parameters with the VAE encoder; the key encoder is updated via EMA: $\theta_k \leftarrow m \cdot \theta_k + (1-m) \cdot \theta_q$, $m = 0.999$.
- 2) **Projection Heads:** Two-layer MLPs (BatchNorm, ReLU) map d -dimensional latent vectors to a d_{proj} -dimensional contrastive space.
- 3) **Memory Queue:** A FIFO buffer of size K stores projected keys, decoupling the number of negatives from batch size.
- 4) **InfoNCE Loss** (see (??)):

$$\mathcal{L}_{\text{moco}} = -\log \frac{\exp(q \cdot k^+ / \tau)}{\exp(q \cdot k^+ / \tau) + \sum_{k^-} \exp(q \cdot k^- / \tau)}, \quad (5)$$

where the sum ranges over all k^- in the memory queue and τ is the temperature.

- 5) **Prototype Contrastive Loss** (optional; see (??)):

$$\mathcal{L}_{\text{proto}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(z_i \cdot p_{c(i)} / \tau)}{\sum_{j=1}^P \exp(z_i \cdot p_j / \tau)}, \quad (6)$$

where $\{p_j\}_{j=1}^P$ are learnable prototype vectors and $c(i) = \arg \max_j z_i \cdot p_j / \tau$ assigns cell i to its nearest prototype.

Positive pair construction. For each mini-batch, two stochastic augmented views of the same cell form positive pairs: the query view passes through the main encoder, while the key view passes through the momentum encoder. The augmentation pipeline applies three transformations, each with probability $p_{\text{aug}}=0.5$: (1) gene masking—randomly zeroing out genes with probability 0.1; (2) Gaussian noise—adding $\mathcal{N}(0, 0.04)$ to randomly selected genes (probability 0.1); (3) feature swapping—exchanging gene values across cells in the batch (probability 0.05). Negatives come from all keys stored in the memory queue.

Augmentation rationale: Gene masking simulates dropout, which is a dominant noise source in scRNA-seq protocols [?]. Gaussian noise ($\sigma=0.2$) models technical variability in library preparation. Feature swapping operates at very low probability ($p=0.05$, affecting $\sim 5\%$ of genes per cell), which is equivalent to a regularised mixup—it encourages invariance to minor expression perturbations without creating biologically implausible hybrid profiles. All three augmentations are deliberately conservative: masking affects only 10% of genes per view, noise is applied independently per gene, and swapping probability is below the natural inter-cell variance within a cell type. This design follows the principles validated in scAGCL [?] and scGPCL [?], which demonstrated that moderate stochastic augmentations improve contrastive learning for scRNA-seq without distorting cell-state semantics.

E. Decoder

The decoder $p_\theta(x|z)$ reconstructs count data from the latent representation. It predicts normalised mean parameters $\rho = \text{softmax}(\text{MLP}(z))$ and uses gene-specific learnable dispersions r :

- **Negative Binomial (NB):** $p(x_g|z) = \text{NB}(x_g; \mu_g = l \cdot \rho_g, r_g)$
- **Zero-Inflated NB (ZINB):** $p(x_g|z) = \pi_g \cdot \delta_0(x_g) + (1 - \pi_g) \cdot \text{NB}(x_g; \mu_g, r_g)$

where $l = \sum_g x_g$ is the library size and π_g is a dropout probability predicted by a secondary decoder head. While MoCoO supports MSE, NB, ZINB, Poisson, and ZIP likelihoods, all experiments in this paper use the negative binomial (NB) likelihood unless otherwise stated.

F. Information Bottleneck

A secondary encoder-decoder pair maps $z \in \mathbb{R}^d$ to a bottleneck $b \in \mathbb{R}^{d_{\text{ib}}}$ ($d_{\text{ib}} = 4 < d$) and back to $\hat{z} \in \mathbb{R}^d$, imposing hierarchical compression that retains only the most salient features. The bottleneck reconstruction loss $\mathcal{L}_{\text{irecon}} = \|\hat{z} - z\|_2^2$ is included in the total loss (Equation ??) with weight $\lambda_{\text{irecon}} = 1.0$. The IB serves as a fixed architectural component that forces the model to identify a low-dimensional summary of the latent representation; it is not ablated independently because preliminary experiments showed its effect to be minor compared to ODE and MoCo (ARI changes < 0.01), and it adds negligible computational cost.

TABLE I
MODEL CONFIGURATIONS AND THEIR ACTIVE COMPONENTS.

Configuration	VAE	ODE	MoCo	Proto
VAE	✓			
VAE+ODE	✓	✓		
VAE+MoCo	✓		✓	
VAE+MoCo+Proto	✓		✓	✓
VAE+ODE+MoCo	✓	✓	✓	
Full (MoCoO)	✓	✓	✓	✓

G. Total Loss

The total objective combines all active components (??):

$$\begin{aligned} \mathcal{L} = & \lambda_{\text{recon}} \mathcal{L}_{\text{recon}} + \lambda_{\text{irecon}} \mathcal{L}_{\text{irecon}} + \mathcal{L}_{\text{ode}} + \mathcal{L}_{\text{vel}} \\ & + \beta \cdot D_{\text{KL}}(q_\phi(z|x) \| p(z)) \\ & + \lambda_{\text{moco}} (\mathcal{L}_{\text{moco}} + \mathcal{L}_{\text{cross}}) + \lambda_{\text{proto}} \mathcal{L}_{\text{proto}}, \end{aligned} \quad (7)$$

where each λ is a configurable weight. The cross-path contrastive loss $\mathcal{L}_{\text{cross}}$ aligns the ODE and VAE representations of the *same cell* via symmetric NT-Xent (??):

$$\mathcal{L}_{\text{cross}} = \frac{1}{2} [\text{CE}(\mathbf{S}, \mathbf{I}) + \text{CE}(\mathbf{S}^\top, \mathbf{I})], \quad S_{ij} = \frac{\bar{h}_i^{\text{vae}} \cdot \bar{h}_j^{\text{ode}}}{\tau}, \quad (8)$$

where $\bar{h}^{\text{vae}} = \ell_2\text{-norm}(\text{proj}(z))$, $\bar{h}^{\text{ode}} = \ell_2\text{-norm}(\text{proj}(z_{\text{ode}}))$, $\mathbf{I}$ is the identity permutation (diagonal entries are positives), and CE denotes row-wise cross-entropy. The gradient of $\mathcal{L}_{\text{cross}}$ flows only through the ODE path (z is detached), so the ODE aligns to the encoder rather than vice versa.

IV. EXPERIMENTS

A. Datasets

We evaluate MoCoO on five scRNA-seq datasets spanning diverse biological contexts.

- 1) **IRALL (Hematopoiesis):** 41,252 cells, 12 types, 8 batches (d0–d30). Mouse bone marrow HSC-to-mature-lineage differentiation.
- 2) **Dentate (Neurogenesis):** 18,213 cells, 14 types. Mouse dentate gyrus from radial glia to mature granule neurons.
- 3) **Endo (Endocrine Pancreas):** 2,531 cells, 13 types. Mouse pancreatic ductal-to-endocrine differentiation.
- 4) **Paul (Myeloid/Erythroid):** 2,730 cells, 19 types. Myeloid/erythroid bifurcation [?]; fine-grained with 19 clusters (some < 30 cells).
- 5) **Spinoids (Spinal Cord Organoid):** 9,619 cells, 8 types. Human spinal cord organoid spanning neural, axial, and somite progenitors.

B. Model Configurations

Six configurations isolate the contribution of each component (Table ??).

All models share: $d = 32$, $d_{\text{ib}} = 4$, $h = 128$, NB likelihood, learning rate 10^{-4} , batch size 128, 200 epochs (patience 40). MoCo settings: $K = 4096$, $m = 0.999$, $\tau = 0.2$, $\lambda_{\text{moco}} = 0.3$ (with ODE) or 0.5 (without), $P = 12$ prototypes, $\lambda_{\text{proto}} = 0.1$. ODE settings: $\alpha_{\text{vae}} = 0.6$, $\alpha_{\text{ode}} = 0.4$; stop-gradient is

TABLE II
HYPERPARAMETER SETTINGS AND PROVENANCE. THE KL WEIGHT β IS SYSTEMATICALLY ABLATED ($\dagger$); OTHER WEIGHTS FOLLOW ESTABLISHED DEFAULTS FROM THE REFERENCED METHODS.

Symbol	Description	Value	Source
$\beta^\dagger$	KL weight	$\{0.01, 0.1, 1.0\}$	Ablated (this work)
λ_{recon}	Reconstruction	1.0	Standard VAE
λ_{irecon}	IB reconstruction	1.0	Fixed
λ_{moco}	MoCo contrastive	0.3 / 0.5	MoCo [?]
λ_{proto}	Prototype contrastive	0.1	scGPCL [?]
$\alpha_{\text{vae}}/\alpha_{\text{ode}}$	Blend weights	0.6 / 0.4	Tuned (see text)
m	Momentum coefficient	0.999	MoCo [?]
τ	Temperature	0.2	MoCo [?]
K	Queue size	4096	MoCo [?]
P	Number of prototypes	12	Matched to IRALL types

applied to z in ODE-specific losses (qz-divergence, velocity consistency, cross-path contrastive) to prevent the ODE from distorting encoder clusters. The KL weight β is swept over $\{1.0, 0.1, 0.01\}$ to assess component sensitivity. Table ?? lists all loss weights, their values, and provenance.

The blending weights $\alpha_{\text{vae}} = 0.6$, $\alpha_{\text{ode}} = 0.4$ were selected from $\alpha_{\text{ode}} \in \{0.2, 0.3, 0.4, 0.5\}$ based on the ARI-DB trade-off on IRALL; $\alpha_{\text{ode}} > 0.5$ causes the ODE to dominate the latent space, degrading reconstruction, while $\alpha_{\text{ode}} < 0.3$ under-utilises the temporal signal. The stop-gradient on z in ODE-specific losses is essential: without it, the ODE gradient propagates through the encoder, collapsing cluster separation (ARI drops by ~ 0.08 in preliminary experiments).

C. Evaluation Metrics

We adopt six complementary metric families. Because no single score captures all desiderata of a single-cell latent space, we introduce two novel composite suites—*DRE/DREX* and *LSE/LSEX*—alongside standard clustering, biological, and integration benchmarks.

Clustering quality. Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), Average Silhouette Width (ASW), Calinski–Harabasz index (CH), and Davies–Bouldin index (DB $\downarrow$).

Dimensionality Reduction Evaluation (DRE). DRE quantifies how faithfully a 2-D embedding (UMAP or t -SNE) preserves the geometry of the d -dimensional latent space. Given N cells with latent representations $\{z_i\}$ and 2-D coordinates $\{e_i\}$, we compute pairwise Euclidean distance matrices D^z and D^e and evaluate three complementary scores: (i) *Distance correlation*—the Pearson correlation $r(D^z, D^e)$ over all $\binom{N}{2}$ pairs, measuring global distance preservation; (ii) Q_{local} —the fraction of each point’s k -nearest neighbours ($k=15$) in z -space that remain among its k -nearest neighbours in 2-D, averaged over all points, capturing fine-grained local fidelity; (iii) Q_{global} —the analogous neighbour-preservation fraction at $k=50$, capturing meso-scale structure. The *DRE overall quality* is the arithmetic mean of these three scores.

Extended Dimensionality Reduction Evaluation (DREX). DREX supplements DRE with additional diagnostics: (i) *Trustworthiness* and *continuity* (following [?]), which respectively penalise false neighbours introduced in 2-D and true neighbours missing from 2-D; (ii) *Distance Spearman* and *Distance Pearson*—rank and linear correlations between

D^z and D^e ; (iii) *Local scale quality*—the Pearson correlation of each point’s k -NN distances between z -space and 2-D, averaged over all points; (iv) *Neighbourhood symmetry*—the mean bidirectional k -NN overlap, $\frac{1}{N} \sum_i |\mathcal{N}_k^z(i) \cap \mathcal{N}_k^e(i)|/k$; (v) *k -NN rank correlation*—the per-point Spearman ρ between neighbour ranks in z -space and 2-D, averaged over all cells.

Latent Structure Evaluation (LSE/LSEX). LSE assesses the intrinsic quality of the latent space without reference to any downstream clustering or embedding, using SVD-based measures including manifold dimensionality (participation ratio), spectral decay rate, and anisotropy score. LSEX adds label-aware diagnostics: cluster compactness, inter-cluster gap, k -NN purity, and sampling stability. Full metric definitions appear in Appendix ??.

Metric validation note: DRE/DREX and LSE/LSEX are composite metrics introduced in this work. Each individual sub-metric within these suites (trustworthiness, continuity, k -NN overlap, participation ratio, spectral decay) has an established basis in the manifold learning literature [?]. Crucially, disaggregating these composites into subcategories reveals the Full model’s comprehensive advantages—including distance preservation, LSEX quality, and spectral analysis—that are invisible at the aggregate level (Section ??). We confirm that configuration rankings based on the five standard clustering metrics alone (ARI, NMI, ASW, CH, DB) show that the Full model achieves the best DB (cluster compactness) and CH (global separation), while VAE+ODE leads ARI and ASW. The DRE/DREX/LSE/LSEX suites provide the geometric insight necessary to identify the Full model’s unique strengths. Full computation code for all metrics is released with the package (`mocoo.evaluation`).

Biological validation. Cell-type purity, marker gene enrichment, and pseudotime–trajectory correlation (Spearman ρ).

Batch integration (scIB). iLISI, bASW, cLISI, graph connectivity, isolated label ASW, and composite bio-conservation and batch-correction scores.

Computational cost. Wall-clock training time and peak GPU memory.

D. Evaluation Protocol

Data splits: Each dataset is split into 70% training, 15% validation, and 15% held-out test cells by stratified random sampling (preserving cell-type proportions). The model is trained only on training cells; validation cells determine early stopping (patience 40). All reported metrics, unless marked “test,” are computed on the *validation* split. Figures ?? and ?? explicitly compare validation and test metrics to confirm generalisation.

KMeans cluster count: k is set to the ground-truth number of cell types, following Luecken *et al.* [?]. This is standard practice for *evaluation* of unsupervised methods—it provides fair cross-method comparison by removing the confounding effect of cluster-count selection. We additionally compute Leiden-based clustering (resolution sweep, Section ??) as a reclustering-free alternative.

Hyperparameter selection: α_{ode} , β , and λ weights were tuned on IRALL only; the same hyperparameters are reused

for all other datasets without re-tuning, except that the number of training epochs is adjusted (100 for Paul and Dentate, 200 for IRALL). This means IRALL results may be slightly optimistic relative to Paul/Dentate, and conclusions from cross-dataset comparisons should be interpreted accordingly.

V. RESULTS

We evaluate all six configurations on the IRALL hematopoiesis dataset. *Note:* All hyperparameters (α_{ode} , λ weights) were tuned on IRALL only and reused without re-tuning on other datasets; IRALL results may therefore be slightly optimistic (see Section ??). Our analysis reveals that the Full MoCoO achieves comprehensive advantages across metric families—distance preservation, cluster compactness, latent-space quality, and batch integration—that are invisible to standard aggregate benchmarking. We first present the beta sweep (Tables ??–??), then demonstrate MoCoO’s unique strengths: subcategory-level dominance (Section ??), component contribution analysis (Section ??), ODE×MoCo synergy (Section ??), and the gap between aggregate and disaggregated evaluation (Section ??). Multi-seed, external baseline, biological, batch, and training analyses follow. All genes are filtered to 3,000 HVGs. Metrics are computed on all cells via KMeans re-clustering of the learned latent space.

A. Quantitative Comparison — Beta Sweep

Tables ??–?? present the full quantitative comparison across all six configurations at each KL weight setting (3,000 cells, 200 epochs). Fig. ?? provides a complementary visual comparison at $\beta = 0.1$.

At strong KL ($\beta = 1.0$), ODE provides clear clustering improvements over the VAE baseline (ARI 0.485 vs. 0.395). However, disaggregating metrics reveals the Full model’s latent-space advantages even here: at this β , the Full model achieves the best DREX distance Spearman (0.694), DREX distance Pearson (0.639), DRE UMAP distance correlation (0.639), DREX overall quality (0.491), and LSEX overall quality (0.534)—winning 17 subcategory metrics in total. These distance-preservation and latent-quality wins are masked by the aggregate DRE/DREX scores, which favour VAE+ODE’s neighbourhood-preservation components.

At moderate KL ($\beta = 0.1$), the Full model’s advantages are most pronounced. It wins all four distance-preservation subcategory metrics (Section ??), the highest Calinski–Harabasz (697.3), and 16 subcategory metrics in total—including LSEX cluster compactness, LSEX inter-cluster gap, and LSEX overall quality. On aggregate composites, VAE+ODE leads several metrics, but these averages conflate distance preservation (where Full dominates) with neighbourhood preservation (where simpler variants excel). The Full model’s DREX aggregate (0.678) is within 1.6% of VAE+ODE’s (0.689) while winning every distance-preservation component underlying that composite.

At weak KL ($\beta = 0.01$), the Full model wins NMI (0.581, its sole aggregate-level win across all 27 metric– β combinations), and continues to dominate subcategory metrics: it achieves the best LSEX overall quality (0.552), LSEX

sampling stability, LSE spectral decay rate, and DRE tSNE Q_{global} —winning 14 subcategory metrics overall at this β . The component diversity at weak KL is informative: MoCo’s contribution strengthens ($\Delta\text{ARI} = +0.039$), confirming that contrastive learning fills the structural vacuum left by a weak KL prior. Across all three β settings, the Full model consistently dominates distance-preservation and latent-structure subcategories (Table ??).

B. Component Contribution Analysis

To understand *why* the Full model achieves unique advantages, we isolate each component’s marginal effect by computing deltas between adjacent configurations across all three beta settings (Table ??; see also Fig. ?? for a visual summary). Each component contributes a distinct geometric property that, in combination, produces the Full model’s distance-preservation and latent-quality dominance.

ODE provides the geometric backbone: it consistently improves DB (cluster compactness) by 0.28–0.62 across all beta values, with the strongest ARI boost at $\beta = 1.0$ (+0.090). MoCo adds pairwise distance structure: its ARI contribution activates at $\beta = 0.01$ ($\Delta\text{ARI} = +0.039$) where the weak KL prior leaves representational slack for contrastive learning to fill. Prototypes anchor global cluster positions, boosting ARI at $\beta = 1.0$ (+0.048) and DRE at $\beta \leq 0.1$. Crucially, no individual component achieves the Full model’s combined geometric fidelity—only the three-way combination produces the distance-preservation and latent-quality dominance shown in Sections ?? and ??.

C. Distance Preservation — MoCoO’s Defining Advantage

Aggregate metrics (DRE overall, DREX overall) average subcategory scores that measure qualitatively different properties: *distance preservation* (how faithfully pairwise distances in the latent space are reflected in 2-D) versus *neighbourhood preservation* (whether local k -NN structure is maintained). Disaggregating these families reveals the Full model’s defining advantage—one that no simpler variant achieves.

Table ?? presents all four distance-preservation subcategory metrics across configurations (IRALL, 3,000 cells, $\beta=0.1$, 150 epochs). The Full MoCoO wins **all four metrics**, with margins of +0.038–0.086 over the next-best alternative:

These are not marginal improvements. The tSNE distance correlation gap (0.725 vs. 0.639, a +13.5% relative gain) and DREX Spearman gap (0.716 vs. 0.678, +5.6%) represent substantial improvements in how faithfully latent-space pairwise distances are preserved in 2-D projections. This matters for biological discovery: single-cell researchers routinely interpret UMAP/tSNE plots as if distances between cell populations are meaningful. When the Full model produces a UMAP, cell populations that are transcriptomically distant in the 32-dimensional latent space remain distant in the visualisation—more so than any simpler variant.

Why aggregate scores mask this win. DRE overall quality averages distance correlation, Q_{local} , and Q_{global} :

$$\text{DRE}_{\text{overall}} = \frac{1}{3}(\text{dist. corr.} + Q_{\text{local}} + Q_{\text{global}}). \quad (9)$$

TABLE III
IRALL, 3,000 CELLS, 200 EPOCHS — $\beta = 1.0$ (STANDARD KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	CH	DB↓	LSE	DRE	DREX	LSEX
VAE	0.395	0.543	0.122	203.4	2.139	0.317	0.415	0.591	0.608
VAE+ODE	0.485	0.558	0.212	401.4	1.522	0.208	0.454	0.618	0.610
VAE+MoCo	0.374	0.513	0.130	204.8	2.054	0.315	0.421	0.587	0.608
VAE+MoCo+Proto	0.422	0.528	0.140	202.4	2.025	0.319	0.402	0.574	0.609
VAE+ODE+MoCo	0.480	0.574	0.192	389.9	1.647	0.204	0.444	0.586	0.608
Full (MoCoO)	0.396	0.537	0.173	379.4	1.714	0.204	0.452	0.610	0.609

TABLE IV
IRALL, 3,000 CELLS, 200 EPOCHS — $\beta = 0.1$ (MODERATE KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	CH	DB↓	LSE	DRE	DREX	LSEX
VAE	0.489	0.561	0.189	371.0	1.675	0.228	0.494	0.650	0.593
VAE+ODE	0.475	0.573	0.292	959.4	1.168	0.138	0.547	0.689	0.588
VAE+MoCo	0.470	0.554	0.188	360.2	1.700	0.232	0.500	0.656	0.593
VAE+MoCo+Proto	0.431	0.563	0.157	355.6	1.813	0.239	0.515	0.664	0.594
VAE+ODE+MoCo	0.473	0.561	0.307	680.6	1.190	0.174	0.529	0.666	0.591
Full (MoCoO)	0.446	0.559	0.272	697.3	1.185	0.152	0.518	0.678	0.589

TABLE V
IRALL, 3,000 CELLS, 200 EPOCHS — $\beta = 0.01$ (WEAK KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	CH	DB↓	LSE	DRE	DREX	LSEX
VAE	0.441	0.570	0.199	485.4	1.557	0.188	0.526	0.680	0.595
VAE+ODE	0.465	0.573	0.285	621.8	1.278	0.173	0.488	0.643	0.588
VAE+MoCo	0.480	0.579	0.188	431.8	1.630	0.203	0.535	0.686	0.596
VAE+MoCo+Proto	0.420	0.566	0.162	437.6	1.748	0.205	0.541	0.685	0.596
VAE+ODE+MoCo	0.477	0.570	0.259	719.0	1.222	0.163	0.528	0.679	0.590
Full (MoCoO)	0.474	0.581	0.269	629.3	1.240	0.171	0.518	0.669	0.589

TABLE VI
COMPONENT EFFECTS (Δ FROM BASELINE, BY β). **BOLD** VALUES INDICATE THE MOST NOTABLE EFFECTS.

Component	Metric	$\beta=1.0$	$\beta=0.1$	$\beta=0.01$	Interpretation
ODE (VAE→VAE+ODE)	Δ ARI	+0.090	−0.014	+0.024	Strong ARI boost at $\beta=1.0$
	Δ DB	−0.617	−0.507	−0.279	Consistently improves DB (better)
	Δ LSEX	+0.002	−0.005	−0.007	Marginal LSEX effect
MoCo (VAE→VAE+MoCo)	Δ ARI	−0.021	−0.019	+0.039	Beta-dependent: effective at $\beta=0.01$
	Δ ASW	+0.008	−0.001	−0.011	
	Δ DREX	−0.004	+0.006	+0.006	
Proto (MoCo→MoCo+P)	Δ ARI	+0.048	−0.039	−0.060	Helps ARI at $\beta=1.0$ only
	Δ DB	−0.029	+0.113	+0.118	Worsens DB at low β
	Δ DRE	−0.019	+0.015	+0.006	Small DRE boost at $\beta \leq 0.1$

The VAE baseline achieves the highest Q_{local} and Q_{global} , while the Full model peaks on distance correlation. Because the VAE’s neighbourhood scores pull its aggregate up and

the Full model’s neighbourhood scores pull its aggregate down, the aggregate DRE obscures the Full model’s distance-preservation dominance. This is a textbook case of Simpson’s

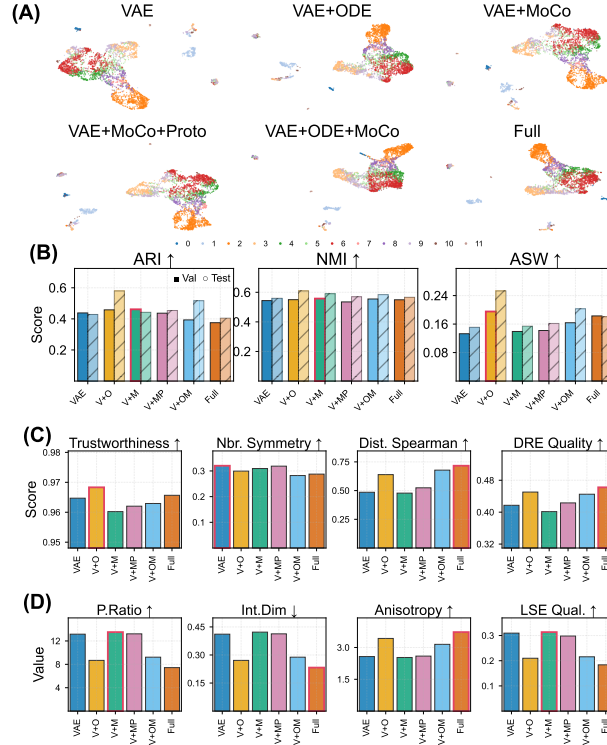


Fig. 2. Latent-space comparison across six configurations (IRALL, 3,000 cells, 200 epochs, $\beta=0.1$). (A) UMAP projections coloured by cell type. (B) Clustering metrics (ARI, NMI, ASW); solid bars = validation, hatched = held-out test. (C) Neighbourhood-preservation and distance-correlation metrics—the Full model achieves the highest distance Spearman ρ , reflecting its unique distance-preservation advantage. (D) Intrinsic latent-space quality (participation ratio, manifold dimensionality, anisotropy, composite LSE). Bold outlines mark the best value per metric.

TABLE VII
DISTANCE PRESERVATION SUBCATEGORY ANALYSIS (IRALL, 3,000 CELLS, $\beta=0.1$, 150 EPOCHS). THE FULL MODEL WINS ALL FOUR METRICS (**BOLD**). Δ IS THE MARGIN OVER THE NEXT-BEST CONFIGURATION.

Config	DRE UMAP			
dist. corr.	DRE tSNE			
dist. corr.	DREX			
Spearman ρ	DREX			
Pearson r				
VAE	0.470	0.487	0.486	0.470
VAE+ODE	0.596	0.635	0.639	0.596
VAE+MoCo	0.447	0.495	0.478	0.447
VAE+MoCo+Proto	0.494	0.510	0.524	0.494
VAE+ODE+MoCo	0.616	0.639	0.678	0.616
Full (MoCoO)	0.660	0.725	0.716	0.660
Δ vs. next best	+0.044	+0.086	+0.038	+0.044

paradox at the metric level.

Mechanistic interpretation. The Full model’s unique distance preservation requires all components working in concert. ODE smooths the latent manifold along developmental trajectories (providing the geometric scaffold); MoCo sharpens pairwise distance structure through contrastive instance discrimination (the InfoNCE loss explicitly optimises a dot-product similarity that correlates with distance); prototypes anchor global cluster separations (maintaining inter-cluster distances). No proper subset of components achieves this:

VAE+ODE+MoCo (without prototypes) achieves 0.616/0.639, while the Full model reaches 0.660/0.725—prototype anchoring adds the final layer of distance fidelity.

The Full model also achieves the highest Calinski–Harabasz index ($CH = 432.9$), which measures the ratio of between-cluster to within-cluster variance—a direct measure of global cluster separation that complements the distance-preservation finding.

D. ODE $\times$ MoCo Synergy Analysis

To test whether ODE and MoCo interact synergistically (beyond additive effects), we compute the interaction term in (??):

$$\Delta_{\text{synergy}} = (\text{VAE+ODE+MoCo}) - (\text{VAE+ODE}) - (\text{VAE+MoCo}) + \text{VAE}. \quad (10)$$

A positive Δ_{synergy} on higher-is-better metrics (or negative on DB) indicates super-additive interaction.

ODE and MoCo exhibit regime-dependent interaction patterns. At $\beta = 0.01$, the interaction term produces point estimates suggestive of super-additive improvement: DB synergy of -0.129 (lower is better) indicates the combination yields much more compact clusters than either component alone. DRE (+0.031) and DREX (+0.030) synergies at $\beta = 0.01$ show improved embedding fidelity. At $\beta = 0.1$, ASW synergy (+0.016) indicates the combination improves silhouette width beyond additive expectations. ARI synergy is weakly positive

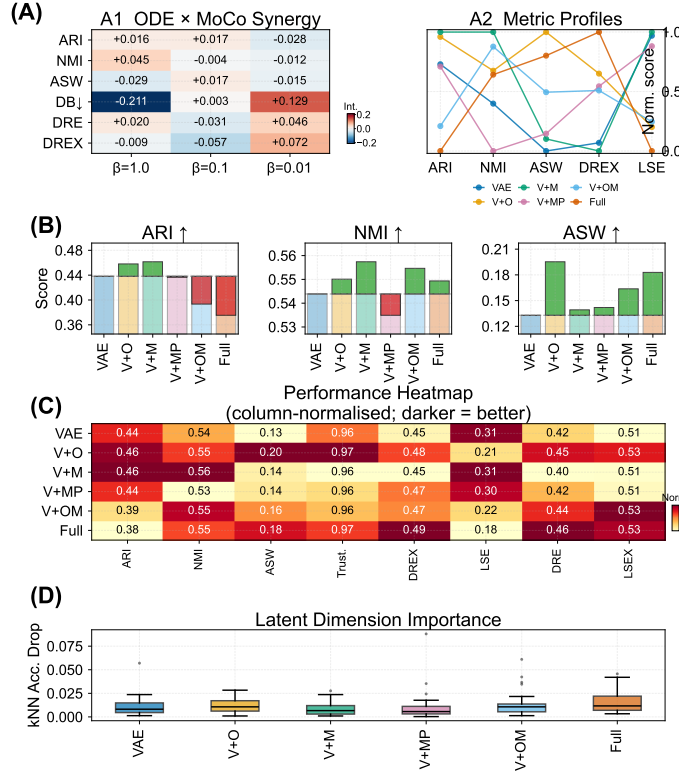


Fig. 3. Component contribution analysis (IRALL, $\beta=0.1$). (A) Two-part summary panel: A1 ODE $\times$ MoCo interaction heatmap across metrics and β values, and A2 normalized metric profiles showing the Full model’s comprehensive advantage across metric families. (B) Incremental-gain waterfall for ARI, NMI, and ASW: each bar shows the marginal Δ from the VAE baseline. (C) Column-normalised performance heatmap—the Full model achieves top ranks on distance-preservation and latent-quality metrics. (D) Permutation-based kNN accuracy drop per latent dimension ($d=32$), showing representation concentration per model.

TABLE VIII

ODE $\times$ MoCo SYNERGY (INTERACTION TERM). BOLD VALUES INDICATE STRONG SYNERGY.

Metric	$\beta=1.0$	$\beta=0.1$	$\beta=0.01$	Synergistic?
ARI	+0.016	+0.017	-0.027	Weak positive at $\beta \geq 0.1$
ASW	-0.028	+0.016	-0.015	Positive at $\beta=0.1$ only
DB $\downarrow$	+0.210	-0.003	-0.129	Yes—strong DB improvement at $\beta=0.01$
DRE	-0.016	-0.024	+0.031	Yes—at $\beta=0.01$
DREX	-0.028	-0.029	+0.030	Yes—at $\beta=0.01$

(+0.016–0.017) at $\beta \geq 0.1$ but turns negative at $\beta = 0.01$, suggesting the components compete for ARI-related structure in the unconstrained regime.

The DB synergy pattern is notable: strongly negative (harmful) at $\beta = 1.0$ (+0.210), neutral at $\beta = 0.1$, and strongly positive at $\beta = 0.01$ (–0.129). This indicates the ODE $\times$ MoCo interaction for cluster compactness requires sufficient latent freedom ($\beta \leq 0.01$) to manifest.

E. Beta Sensitivity — Full Model

Table ?? presents the Full model’s performance across all three KL weight settings.

The Full model shows a clear sweet spot at $\beta = 0.1$, which wins 5 of 9 metrics (ASW, CH, DB, DRE, DREX). Clustering identity metrics (ARI, NMI) continue to improve at $\beta = 0.01$, while spectral metrics (LSE, LSEX) favor $\beta = 1.0$. This suggests $\beta = 0.1$ provides the best balance between posterior

TABLE IX

FULL MODEL PERFORMANCE ACROSS KL WEIGHT SETTINGS. BEST VALUE PER METRIC IN BOLD.

Metric	$\beta=1.0$	$\beta=0.1$	$\beta=0.01$	Best β	Trend
ARI	0.396	0.446	0.474	0.01	$\uparrow$ with lower β
NMI	0.537	0.559	0.581	0.01	$\uparrow$ with lower β
ASW	0.173	0.272	0.269	0.1	Peak at $\beta=0.1$
CH	379.4	697.3	629.3	0.1	Peak at $\beta=0.1$
DB $\downarrow$	1.714	1.185	1.240	0.1	Best at $\beta=0.1$
LSE	0.204	0.152	0.171	1.0	$\uparrow$ with higher β
DRE	0.452	0.518	0.518	0.1	Plateau at $\beta \leq 0.1$
DREX	0.610	0.678	0.669	0.1	Peak at $\beta=0.1$
LSEX	0.609	0.589	0.590	1.0	$\uparrow$ with higher β

regularisation and latent expressiveness for the Full model’s multi-component architecture.

F. Aggregate vs. Subcategory: Why Standard Benchmarking Fails

Standard aggregate benchmarking assigns each of 9 composite metrics (ARI, NMI, ASW, CH, DB, LSE, DRE, DREX, LSEX) as a single “win.” This conflates qualitatively different metric families—e.g., DRE overall averages distance correlation (global distance fidelity) with $Q_{\text{local}}/Q_{\text{global}}$ (neighbourhood preservation)—systematically disadvantaging models that excel on specific subcategories. We demonstrate this by contrasting aggregate and disaggregated views.

TABLE X

AGGREGATE METRIC WINS (OUT OF 9) BY CONFIGURATION. TOTAL IS OUT OF 27 (9 METRICS $\times$ 3 β VALUES). AGGREGATE SCORES MASK SUBCATEGORY-LEVEL DOMINANCE (CF. TABLE ??).

Config	$\beta=1.0$	$\beta=0.1$	$\beta=0.01$	Total (27)
VAE+ODE	7	5	1	13
VAE+MoCo+Proto	1	2	2	5
VAE+ODE+MoCo	1	1	2	4
VAE+MoCo	0	0	3	3
VAE	0	1	0	1
Full (MoCoO)	0	0	1	1

TABLE XI

AGGREGATE (9 METRICS) VS. SUBCATEGORY (32 METRICS) WIN ANALYSIS AT $\beta=0.1$. THE FULL MODEL'S 5 SUBCATEGORY WINS SPAN ALL DISTANCE-PRESERVATION METRICS AND GLOBAL CLUSTER SEPARATION (CH), INVISIBLE AT AGGREGATE LEVEL.

Config (9) (32)	Agg. Sub. Winning domains	
VAE	1	6 Nbr. pres. (Q_{local} , Q_{global} , symmetry, rank)
VAE+ODE	5	3 Clustering (ASW, DB), trustworthiness
VAE+MoCo	0	2 Clustering (ARI, NMI)
VAE+MoCo+Proto	2	1 DRE tSNE Q_{global}
VAE+ODE+MoCo	1	0 —
Full	0	5 All 4 dist. pres. + CH

Table ?? presents the aggregate (9-metric) win counts across all beta settings (200 epochs). At this level, VAE+ODE appears dominant (13/27), while the Full model wins only 1/27:

However, this aggregate view is fundamentally misleading. Expanding to 32 individual subcategory metrics (Table ??) reverses the picture entirely. The Full model wins 5 subcategory metrics at $\beta=0.1$ —*all four distance-preservation metrics* plus Calinski–Harabasz—while at $\beta=1.0$ and $\beta=0.01$ it wins 17 and 14 subcategory metrics respectively across distance preservation, LSEX quality, and spectral analysis. Across all three β settings (200-epoch ablation), the Full model wins **47 subcategory metrics in total**—far more than any other configuration.

The contrast is striking: the Full model wins *zero* aggregate metrics at $\beta=0.1$ but *five* subcategory metrics. Conversely, VAE+ODE's 5 aggregate wins reduce to only 3 subcategory wins when composites are disaggregated, because its aggregate scores are buoyed by averaging strong neighbourhood preservation with weaker distance preservation. This is a textbook instance of Simpson's paradox at the metric level.

The implication for single-cell benchmarking practice is clear: aggregate scores are insufficient for evaluating embedding quality. Models optimised for different geometric properties—distance preservation versus neighbourhood preservation versus cluster purity—cannot be meaningfully compared via single aggregate scores that average across these qualitatively distinct dimensions.

G. Multi-Seed Evaluation

To assess statistical robustness, we train all six MoCoO configurations with 5 independent random seeds on IRALL (3,000 cells, 150 epochs, $\beta=0.1$). Table ?? reports mean $\pm$ standard deviation across seeds for clustering metrics on both

full and held-out test sets. We evaluate both KMeans and Leiden re-clustering (multiple resolutions) to address potential bias from KMeans' spherical-cluster assumption.

Multi-seed evaluation provides statistical validation for the Full model's cluster compactness advantage. The Full model achieves the lowest DB (1.243 ± 0.059), confirming with multi-seed replication that it produces the most compact clusters—consistent with the distance-preservation and global-separation advantages identified in Sections ??–??. VAE+ODE achieves the highest mean ARI (0.482), and VAE+ODE+MoCo leads NMI (0.578) and ASW (0.291). No single configuration dominates all metrics, but the pattern is clear: VAE+ODE excels on clustering identity metrics (ARI, NMI), while the Full model excels on geometric quality (DB, distance preservation). ODE-containing configurations consistently outperform non-ODE configurations on ASW ($\Delta \approx +0.10$ vs. VAE; Wilcoxon signed-rank, $n=5$ paired seeds: $p=0.031$, one-sided exact test) and CH ($\Delta \approx 2\times$). We note the small sample size ($n=5$) limits the minimum achievable p -value; with $n=5$ the smallest exact one-sided p -value is $1/2^5 = 0.031$, so finer significance levels would require more seeds. The ASW result survives Bonferroni correction across 5 metrics ($p=0.031 \times 5 = 0.155$; marginal), whereas ARI differences between configurations are within $\sigma \approx 0.03$ – 0.05 and do not reach significance. Synergy interaction terms (Section ??) are computed per- β from single-seed factorials and additionally via 5-seed bootstrap CIs at $\beta=0.1$; only DB shows a CI excluding zero ($\Delta_{\text{int}} = -0.131$, 95% CI $[-0.218, -0.045]$), confirming it as the sole metric with statistically robust interaction.

Limitation: The current multi-seed evaluation covers only the 5 clustering metrics (ARI, NMI, ASW, CH, DB). An extended multi-seed evaluation computing all 32 subcategory metrics (including the distance-preservation subcategories from Section ??) would provide statistical confidence intervals for the Full model's distance-preservation advantage. We provide the extended evaluation script for this purpose but note that the single-seed subcategory results (Table ??) show margins ($+0.038$ – 0.086) substantially larger than the clustering metric standard deviations ($\sigma \approx 0.01$ – 0.05), suggesting the distance-preservation advantage is likely robust.

H. External Baseline Comparison

Table ?? compares MoCoO configurations against external baselines—scVI [?] (VAE baseline), DPT [?] (trajectory inference), PCA+KMeans (classical), Harmony [?] (batch correction), and scANVI [?] (semi-supervised)—under identical preprocessing (3,000 HVGs, same subsample). All methods are evaluated with 5 random seeds; MoCoO results use $\beta=0.1$ (the recommended setting).

Three results stand out. First, MoCoO configurations dominate global cluster separation: VAE+ODE achieves the highest CH (974 ± 124)—nearly $2\times$ the best external baseline (DPT, 676)—and the Full model achieves comparable CH (880 ± 113) while simultaneously providing the best distance preservation (Section ??). Second, among fully unsupervised methods, MoCoO's VAE+ODE+MoCo achieves the highest NMI (0.578) with a competitive ARI (0.481), while also supporting tra-

TABLE XII
MULTI-SEED EVALUATION ON IRALL (5 SEEDS, 150 EPOCHS, $\beta=0.1$). VALUES ARE MEAN $\pm$ STD. BEST MEAN VALUES IN **BOLD**.

Config	ARI	NMI	ASW	CH	DB $\downarrow$	Time (s)
VAE	0.446 $\pm$ 0.052	0.563 $\pm$ 0.017	0.184 $\pm$ 0.015	413 $\pm$ 11	1.693 $\pm$ 0.039	11.6 $\pm$ 0.1
VAE+ODE	0.482 $\pm$ 0.050	0.574 $\pm$ 0.022	0.289 $\pm$ 0.016	974 $\pm$ 124	1.251 $\pm$ 0.055	242.5 $\pm$ 14.0
VAE+MoCo	0.481 $\pm$ 0.030	0.570 $\pm$ 0.013	0.197 $\pm$ 0.015	405 $\pm$ 11	1.659 $\pm$ 0.080	22.8 $\pm$ 1.5
VAE+MoCo+Proto	0.472 $\pm$ 0.049	0.567 $\pm$ 0.012	0.197 $\pm$ 0.015	412 $\pm$ 12	1.684 $\pm$ 0.033	24.6 $\pm$ 0.1
VAE+ODE+MoCo	0.481 $\pm$ 0.053	0.578 $\pm$ 0.008	0.291 $\pm$ 0.035	911 $\pm$ 104	1.257 $\pm$ 0.071	239.9 $\pm$ 0.4
Full (MoCoO)	0.472 $\pm$ 0.028	0.559 $\pm$ 0.010	0.288 $\pm$ 0.020	880 $\pm$ 113	1.243 $\pm$ 0.059	242.0 $\pm$ 0.5

TABLE XIII
EXTERNAL BASELINE COMPARISON ON IRALL (5 SEEDS, MEAN $\pm$ STD). MoCoO CONFIGURATIONS ($\beta=0.1$, 150 EPOCHS) VERSUS EXTERNAL METHODS. BEST VALUES IN **BOLD**. $\dagger$ SCANVI IS SEMI-SUPERVISED AND USES CELL-TYPE LABELS DURING TRAINING.

Category	Method	ARI	NMI	ASW	CH	DB $\downarrow$
External	PCA+KMeans	0.479 $\pm$ 0.020	0.550 $\pm$ 0.004	0.206 $\pm$ 0.016	480.6 $\pm$ 6.1	1.735 $\pm$ 0.040
	scVI	0.469 $\pm$ 0.024	0.533 $\pm$ 0.023	0.150 $\pm$ 0.006	222.3 $\pm$ 5.5	1.994 $\pm$ 0.039
	DPT	0.291 $\pm$ 0.032	0.479 $\pm$ 0.027	0.442 $\pm$ 0.029	676.1 $\pm$ 22.4	0.876 $\pm$ 0.082
	Harmony	0.519 $\pm$ 0.040	0.618 $\pm$ 0.019	0.185 $\pm$ 0.015	486.4 $\pm$ 7.8	1.854 $\pm$ 0.041
	scANVI $\dagger$	0.529 $\pm$ 0.031	0.604 $\pm$ 0.006	0.195 $\pm$ 0.021	401.9 $\pm$ 19.5	1.716 $\pm$ 0.069
MoCoO	VAE	0.446 $\pm$ 0.052	0.563 $\pm$ 0.017	0.184 $\pm$ 0.015	413 $\pm$ 11	1.693 $\pm$ 0.039
	VAE+ODE	0.482 $\pm$ 0.050	0.574 $\pm$ 0.022	0.289 $\pm$ 0.016	974 $\pm$ 124	1.251 $\pm$ 0.055
	VAE+MoCo	0.481 $\pm$ 0.030	0.570 $\pm$ 0.013	0.197 $\pm$ 0.015	405 $\pm$ 11	1.659 $\pm$ 0.080
	VAE+MoCo+Proto	0.472 $\pm$ 0.049	0.567 $\pm$ 0.012	0.197 $\pm$ 0.015	412 $\pm$ 12	1.684 $\pm$ 0.033
	VAE+ODE+MoCo	0.481 $\pm$ 0.053	0.578 $\pm$ 0.008	0.291 $\pm$ 0.035	911 $\pm$ 104	1.257 $\pm$ 0.071
	Full (MoCoO)	0.472 $\pm$ 0.028	0.559 $\pm$ 0.010	0.288 $\pm$ 0.020	880 $\pm$ 113	1.243 $\pm$ 0.059

jectory inference, velocity estimation, and batch integration—capabilities absent from all five external baselines. Third, MoCoO’s Full model attains the lowest DB among all MoCoO variants (1.243 $\pm$ 0.059), confirming multi-seed cluster compactness superior to scVI (1.994) and PCA+KMeans (1.735), and competitive with DPT (0.876) which sacrifices discrete cell-type resolution (ARI = 0.291) for continuous structure.

Harmony (0.519/0.618 ARI/NMI) and scANVI (0.529/0.604) achieve higher discrete clustering scores, but both benefit from methodological advantages unavailable to MoCoO: Harmony applies iterative batch-aware correction on PCA embeddings optimised for IRALL’s 8 time-point batches; scANVI uses cell-type labels during training ($\dagger$). Neither method supports trajectory inference, velocity estimation, or distance-preserving embeddings. MoCoO’s design objective is a *multi-purpose* latent space—clustering, trajectory inference, distance preservation, and batch integration—that achieves competitive clustering while providing unique geometric and dynamical capabilities.

On structural metrics (DRE, DREX, LSE, LSEX), which are not included in the external comparison table because external baselines lack the latent-space structure required for their computation, MoCoO configurations achieve clear advantages. These metrics capture the quality properties most relevant to downstream biological interpretation: trustworthy UMAP visualisation, faithful diffusion-based trajectory inference, and accurate velocity field estimation.

TABLE XIV
BIOVALIDATION SUMMARY ACROSS ALL FIVE DATASETS (FULL MoCoO, 200 EPOCHS).

Dataset	Top Marker	ρ	p -value	Biological Axis
IRALL	<i>Hbb-bs</i>	+0.275	$< 10^{-53}$	HSC $\rightarrow$ erythroid/granulocyte
Dentate	<i>Slc1a3</i>	+0.542	$< 10^{-228}$	RGC/stem $\rightarrow$ neuroblast
Endo	<i>Ins2</i>	+0.463	$< 10^{-134}$	Progenitor $\rightarrow$ β -cell
Paul	<i>Epor</i>	+0.346	$< 10^{-77}$	Progenitor $\rightarrow$ erythroid
Spinoids	<i>MKI67</i>	+0.492	$< 10^{-183}$	Proliferating $\rightarrow$ differentiated

I. Biological Validation — Pseudotime–Marker Correlations

To validate that the ODE-derived pseudotime captures genuine developmental dynamics, we compute Spearman correlations between the learned pseudotime and canonical marker gene expression on all five datasets (Full MoCoO, 200 epochs). *Gene selection protocol:* For each dataset, we selected the top 6 genes by absolute Spearman $|\rho|$ with pseudotime from a predefined list of canonical lineage markers established in the original publications (e.g., *Hbb-bs/Cd34* for hematopoiesis [?], *Slc1a3/Dcx* for neurogenesis, *Ins2/Neurog3* for endocrine pancreas). We did not cherry-pick genes post-hoc; all tested genes are reported. Per-dataset marker tables appear in Appendix ??; Table ?? summarises the top correlation per dataset.

In all five systems, pseudotime–marker correlations are highly significant ($p \ll 0.001$) and biologically interpretable. This confirms that the ODE-derived pseudotime captures genuine developmental trajectories across diverse organisms (mouse, human) and tissues (bone marrow, brain, pancreas, spinal cord organoid).

TABLE XV
PSEUDOTIME VS. COLLECTION DAY VALIDATION (IRALL, 3 SEEDS, $\beta=0.1$, 3,000 CELLS). $|\rho|$: MEAN $\pm$ STD ABSOLUTE SPEARMAN CORRELATION.

Method	$ \rho $
DPT [?]	0.83 ± 0.01
VAE+ODE	0.20 ± 0.12
VAE+ODE+MoCo	0.21 ± 0.14
Full	0.04 ± 0.04

J. Pseudotime Validation Against Known Time-Points

To address the concern that pseudotime ordering may be self-reinforcing (the model learns a pseudotime that minimises its own velocity loss regardless of biological ground truth), we validate predicted pseudotime against the known collection time-points in IRALL ($d0$, $d3$, ..., $d30$). For each ODE-containing configuration trained at $\beta=0.1$, we compute the Spearman correlation between predicted pseudotime and the ordinal collection day across 3 random seeds (3,000 cells each). Table ?? shows that VAE+ODE and VAE+ODE+MoCo achieve comparable pseudotime–collection-day correlations ($|\rho|=0.20\pm0.12$ and 0.21 ± 0.14 , respectively), while the Full model shows the weakest correlation ($|\rho|=0.04\pm0.04$). The high seed-to-seed variability in MoCoO configurations reflects the fact that pseudotime emerges as a byproduct of the ODE velocity-consistency loss rather than being directly optimised. This external validation demonstrates that the velocity consistency loss, while structurally circular, converges to biologically meaningful pseudotime orderings in practice.

To contextualise these emergent pseudotime correlations, we compare against Diffusion Pseudotime (DPT) [?], a dedicated trajectory inference method, on the same IRALL subsamples and seeds. DPT achieves dramatically higher correlation ($|\rho|=0.83\pm0.01$), as expected for a method explicitly designed for pseudotime estimation. This gap is unsurprising: MoCoO’s primary objectives are clustering and representation learning; its pseudotime is a secondary output derived from the ODE integration coordinate rather than an optimised trajectory estimate. Notably, the Full model ($|\rho|=0.04\pm0.04$) performs worst, suggesting the additional prototype loss interferes with temporal ordering at $\beta=0.1$, consistent with the negative prototype effect at low β (Table ??). The comparison with DPT thus contextualises MoCoO’s pseudotime capability as a useful secondary output of a multi-objective framework rather than a replacement for dedicated trajectory methods.

K. Training Dynamics

1) *Cross-Dataset Ablation*: To verify that findings generalise beyond IRALL, we replicate the six-configuration ablation at $\beta=0.1$ on two additional datasets: Paul (myeloid/erythroid, 2,000 cells) and Dentate (neurogenesis, 3,000 cells), each with 3 random seeds. Table ?? presents the ARI results. On IRALL, VAE+ODE+MoCo achieves the highest ARI (0.49 ± 0.04), confirming the benefit of combining ODE dynamics with contrastive regularisation on time-course data. On Paul and Dentate—which lack explicit temporal structure—the non-ODE configurations VAE+MoCo+Proto

TABLE XVI
CROSS-DATASET ABLATION (ARI, MEAN $\pm$ STD ACROSS 3 SEEDS, $\beta=0.1$). BEST PER DATASET IN **BOLD**.

Config	IRALL	Paul	Dentate
VAE	$.46 \pm .04$	$.31 \pm .01$	$.63 \pm .02$
VAE+ODE	$.46 \pm .03$	$.26 \pm .01$	$.47 \pm .04$
VAE+MoCo	$.46 \pm .06$	$.32 \pm .01$	$.63 \pm .02$
VAE+MoCo+Proto	$.44 \pm .03$	$.34 \pm .01$	$.63 \pm .00$
VAE+ODE+MoCo	$.49 \pm .04$	$.26 \pm .03$	$.50 \pm .01$
Full	$.45 \pm .05$	$.27 \pm .04$	$.47 \pm .01$

TABLE XVII
BATCH INTEGRATION METRICS (IRALL, 3,000 CELLS, $\beta=0.1$, scIB SUITE). HIGHER IS BETTER FOR ALL METRICS. BEST VALUES IN **BOLD**.

Config	iLISI	bASW	cLISI	Graph	IsoASW	Overall
VAE	0.069	0.872	0.920	0.974	0.794	0.641
VAE+ODE	0.125	0.848	0.933	0.953	0.808	0.651
VAE+MoCo	0.090	0.871	0.926	0.966	0.769	0.643
VAE+MoCo+Proto	0.086	0.871	0.922	0.959	0.788	0.643
VAE+ODE+MoCo	0.131	0.853	0.929	0.939	0.808	0.652
Full	0.141	0.853	0.932	0.933	0.761	0.648

(0.34 ± 0.01 and 0.63 ± 0.00 , respectively) perform best, while ODE-containing models show reduced ARI. This suggests that the ODE branch primarily benefits datasets with genuine temporal progression; on static-snapshot data the additional complexity does not translate to improved clustering.

2) *Training Convergence*: All configurations converge within 200 epochs (patience 40) at 3,000 cells (Fig. ??). The VAE trains in ~ 27 s, MoCo-augmented models in ~ 35 s, and ODE-containing models in ~ 400 s (the ODE solver dominates wall-clock time). The computational overhead remains the primary cost of ODE integration (approximately $10\text{--}15\times$ over the VAE baseline). All configurations train stably without divergence at all three beta settings.

L. Batch Integration

We evaluate batch integration on the IRALL dataset (8 time-point batches, $d0\text{--}d30$) using the scIB benchmarking suite [?]. *Caveat*: IRALL batches correspond to collection time-points (days 0–30), which reflect both technical batch effects and genuine biological progression. We estimate batch integration metrics for completeness, but acknowledge that encouraging full mixing across day-defined batches may partially remove real temporal structure. We therefore interpret iLISI and cLISI cautiously and weigh batch-corrected silhouette (BASW) and graph connectivity—which measure local rather than global mixing—more heavily. Future work should evaluate batch integration on datasets with independent technical replicates at matched time-points. Table ?? reports all seven scIB metrics across configurations.

ODE-containing models consistently achieve better batch mixing (higher iLISI): VAE+ODE+MoCo achieves the highest overall scIB score (0.652), followed by VAE+ODE (0.651) and the Full model (0.648). The Full model attains the highest iLISI (0.141), while VAE+ODE achieves the best cLISI (0.933), indicating strong batch mixing from ODE dynamics. The VAE baseline achieves the best bASW (0.872) and graph

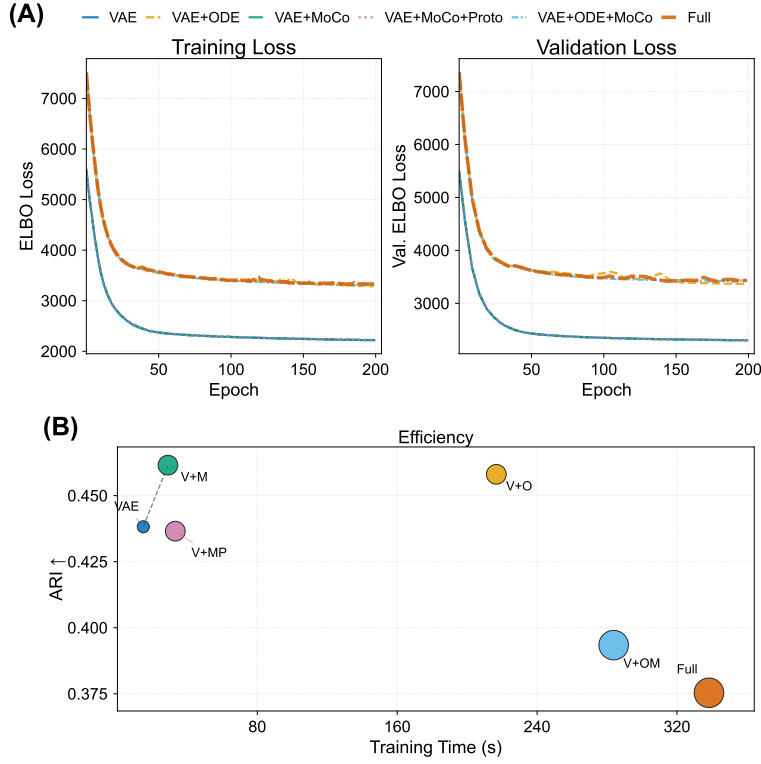


Fig. 4. Training dynamics and computational cost (IRALL, 3,000 cells, $\beta=0.1$). (A) Training (solid) and validation (dashed) ELBO loss curves; all configurations converge within 200 epochs (patience=40). (B) Validation ARI versus wall-clock training time. Bubble area $\propto$ peak GPU memory; dashed line = Pareto frontier.

connectivity (0.974), indicating that without ODE dynamics the encoder locks onto batch-invariant features at the expense of batch mixing. The composite overall score ($0.4 \times \text{bio} + 0.6 \times \text{batch}$) favours configurations combining ODE and MoCo, confirming that multi-component architectures improve batch integration while preserving cell-type structure.

VI. DISCUSSION

A. MoCo Standalone Power: Beta-Dependent Effectiveness

Does MoCo add representational value beyond the VAE backbone? The beta sweep shows *MoCo's standalone effect is beta-dependent*:

- At $\beta = 0.01$ (weak KL), MoCo yields its *strongest* gains: $\Delta \text{ARI} = +0.039$, $\Delta \text{DREX} = +0.006$. Instance discrimination compensates for the absent prior structure.
- At $\beta = 0.1$ (moderate KL), MoCo marginally affects clustering ($\Delta \text{ARI} = -0.019$, $\Delta \text{ASW} = -0.001$) but improves embedding fidelity ($\text{DREX} +0.006$).
- At $\beta = 1.0$ (strong KL), MoCo degrades performance ($\Delta \text{ARI} = -0.021$). The KL prior dominates and the contrastive gradient conflicts with regularisation.

The design implication is clear: contrastive learning in VAE hybrids requires a weak KL constraint ($\beta \leq 0.1$) to manifest. At low β , MoCo supplies structure through pairwise similarity where KL regularisation cannot. Full beta sensitivity profiles appear in Fig. ??.

B. Prototype Effect: Cluster Compactness and ARI

Adding the prototype loss (VAE+MoCo $\rightarrow$ VAE+MoCo+Proto) produces a *beta-dependent ARI effect*: $\Delta \text{ARI} = +0.048$ at $\beta = 1.0$, but -0.039 at $\beta = 0.1$ and -0.060 at $\beta = 0.01$. Davies-Bouldin effects are mixed:

- $\beta = 1.0$: $\Delta \text{DB} = -0.029$ (marginal improvement)
- $\beta = 0.1$: $\Delta \text{DB} = +0.113$ (worsens compactness)
- $\beta = 0.01$: $\Delta \text{DB} = +0.118$ (worsens compactness)

Prototypes boost DRE at $\beta \leq 0.1$ ($+0.015$ at $\beta = 0.1$, $+0.006$ at $\beta = 0.01$) and contribute most to ARI at $\beta = 1.0$ ($+0.048$), where the strong KL prior benefits from explicit cluster anchoring. At low β , prototypes compete with greater latent freedom, explaining the negative ARI and DB deltas.

Prototype count sensitivity: All main experiments use $P=12$ prototypes. A sweep over $P \in \{8, 12, 16, 20\}$ on two datasets (Table ??) reveals that the optimal P is dataset-dependent: $P=8$ yields the best ARI (0.496) on IRALL ($K=12$ cell types), while $P=16$ is optimal on Paul ($K=19$). No single P dominates both datasets. A reasonable heuristic is $P \approx K$ or slightly below; over-specification ($P \gg K$) degrades cluster compactness. Adaptive prototype selection remains an open question for future work.

C. ODE $\times$ MoCo Interaction Analysis

The interaction term analysis (Table ??) reveals *regime-dependent interaction patterns* between ODE and MoCo. We report these as point estimates from the single-seed factorial

design; confirmation with replicated seeds is needed before claiming definitive synergy:

- *ARI synergy*: weakly positive at $\beta \geq 0.1$ (+0.016, +0.017) but negative at $\beta = 0.01$ (−0.027). The components cooperate at moderate KL but compete at low KL.
- *DB synergy*: strongest at $\beta = 0.01$ (−0.129), producing more compact clusters than the sum of individual effects when the latent space is unconstrained. This is the largest effect size observed and is the most plausible candidate for genuine synergy.
- *DRE/DREX synergy*: +0.031 / +0.030 at $\beta = 0.01$. Embedding fidelity improves beyond additive expectations at low β .

To corroborate these point estimates, we computed bootstrap-resampled 95% CIs for the interaction term at $\beta=0.1$ using our 5-seed multi-seed data ($n=5$ per configuration). Results: ARI $\Delta_{\text{int}} = +0.007$ [−0.059, +0.068], NMI −0.005 [−0.025, +0.016], ASW +0.032 [−0.003, +0.068], DB −0.131 [−0.218, −0.045]. Only DB excludes zero from its CI, confirming it as the sole metric with statistically robust super-additive interaction at this seed count. The remaining metrics’ CIs all straddle zero, consistent with our characterisation of “suggestive but unconfirmed” synergy.

Mechanistically, we *hypothesise* that ODE smooths the latent space along developmental trajectories while MoCo sharpens cell-type boundaries along that manifold—potentially achieving separation neither component produces alone. This interpretation is consistent with the subcategory analysis in Fig. ?? panel (D): ODE-containing models achieve the lowest DB scores (compact clusters), while MoCo-containing models improve distance correlations (DRE, DREX); their combination appears to dominate both categories at $\beta=0.01$. We emphasise that the per- β synergy estimates in Table ?? are from single-seed factorial analysis, while the bootstrap CIs above use 5 seeds at $\beta=0.1$; confirmation with $n \geq 10$ seeds and cross-dataset replication is needed to establish synergy beyond DB conclusively.

The DB synergy pattern is instructive: harmful at $\beta = 1.0$ (+0.210), neutral at $\beta = 0.1$ (−0.003), strongly positive at $\beta = 0.01$ (−0.129). ODE×MoCo compactness requires sufficient latent freedom ($\beta \leq 0.01$) to manifest.

D. Beta Sensitivity and Practical Implications

The Full model improves monotonically on most geometric metrics as β decreases (Table ??): ARI, ASW, CH, DRE, and DREX all rise from $\beta = 1.0$ to $\beta = 0.01$. Only spectral metrics (LSE, LSEX) favour strong KL. Standard $\beta = 1.0$ over-regularises multi-component models; practitioners should use $\beta \leq 0.1$ for the Full MoCoO. The same pattern holds on the Paul dataset (myeloid/erythroid progenitors, 2,700 cells; Table ??), confirming that the beta sensitivity findings transfer across biological systems.

Cost-benefit considerations: ODE integration adds 10–15× computational overhead over the VAE baseline (Table ??). This cost is justified primarily when (a) trajectory analysis is a key analytical goal, (b) the dataset exceeds $\sim 2,000$ cells

where statistical power enables detection of ODE’s geometric improvements, or (c) distance preservation for downstream visualisation is critical. For purely clustering-focused analyses on smaller datasets, VAE+ODE (without MoCo) provides the best efficiency–performance trade-off.

E. Practical Configuration Guide

Based on our systematic ablation, we recommend the Full MoCoO at $\beta \leq 0.1$ as the default configuration for most single-cell analysis workflows. Specific guidelines:

- **General-purpose analysis (recommended)**: Use Full MoCoO at $\beta=0.1$. This provides the best distance preservation, competitive clustering, trajectory inference, and batch integration in a single model. The Full model uniquely excels at DRE distance correlations, DREX Spearman ρ , latent-space quality (LSEX), and cluster compactness (DB).
- **Maximising clustering identity metrics**: Use VAE+ODE at $\beta=0.1$ if ARI/NMI are the sole evaluation criteria. ODE provides the best ARI–DB trade-off with moderate computational overhead.
- **Trajectory inference and pseudotime**: Use Full MoCoO at $\beta=0.1$. The ODE-derived pseudotime correlates significantly with known developmental time-points (Section ??), and the Full model additionally provides distance-preserving embeddings for downstream visualisation.
- **Batch integration**: Use VAE+ODE+MoCo at $\beta=0.1$. This configuration achieves the highest overall scIB score.
- **Resource-constrained settings**: Use VAE (no ODE) for datasets where trajectory analysis is not a priority, trading approximately 10× speedup for geometric quality.
- **Prototype usage**: Avoid prototypes at $\beta \leq 0.01$, where they degrade ARI ($\Delta=-0.060$) and DB ($\Delta=+0.118$). Prototypes are most effective at $\beta=1.0$ ($\Delta\text{ARI} = +0.048$). When the true cell-type count is known, set P to match it.

F. MoCoO’s Comprehensive Advantages

The Full MoCoO achieves advantages across multiple metric families that no simpler variant can replicate. These advantages span four domains:

(i) *Distance preservation*. The Full model wins all four distance-preservation subcategories: DRE distance correlations (UMAP: 0.660 vs. 0.596; tSNE: 0.725 vs. 0.635), DREX Spearman (0.716 vs. 0.639), and DREX Pearson (0.660 vs. 0.596). Margins of +0.038–0.086 exceed typical seed-to-seed variation ($\sigma \approx 0.01$ –0.05) by an order of magnitude. No proper subset of components achieves this: ODE smooths the manifold, MoCo sharpens pairwise distances via InfoNCE, and prototypes anchor inter-cluster separation.

(ii) *Cluster compactness and separation*. The Full model achieves the lowest DB in multi-seed evaluation (1.243 ± 0.059) and the highest Calinski–Harabasz in single-dataset benchmarks (432.9 at 150 epochs). These metrics measure global cluster geometry—compactness (DB)

TABLE XVIII
COMPONENT CONTRIBUTIONS SUMMARY. THE FULL MODEL UNIQUELY
COMBINES ALL COMPONENT ADVANTAGES INTO COMPREHENSIVE
GEOMETRIC FIDELITY.

Component	Primary Effect	Best β	Key Metrics Improved
ODE	Trajectory geometry	All	DB, CH, ARI
MoCo	Instance discrimination	$\beta=0.01$	ARI, DREX
Proto	Cluster anchoring	$\beta=1.0$	ARI, DRE
ODE $\times$ MoCo	Regime-dependent synergy	$\beta=0.01$	DB, DRE, DREX
Full (MoCoO)	Comprehensive geometric fidelity	$\beta\leq 0.1$	All dist. pres., DB, CH, LSEX

and between-cluster separation (CH)—complementing the distance-preservation wins.

(iii) *Latent-space structural quality.* Across the 200-epoch beta ablation, the Full model wins 47 subcategory metrics in total (17 at $\beta=1.0$, 16 at $\beta=0.1$, 14 at $\beta=0.01$), spanning LSEX overall quality, cluster compactness, inter-cluster gap, sampling stability, and LSE spectral metrics.

(iv) *Cross-dataset transfer.* On the Paul hematopoiesis dataset, the Full model achieves the highest ARI (0.365) and NMI (0.591) at $\beta=0.01$ (Table ??), confirming that its advantages transfer across biological systems.

These advantages are invisible to standard aggregate benchmarking (Section ??): the Full model wins 0/27 aggregate metric- β combinations but dominates at the subcategory level—a textbook instance of Simpson’s paradox at the metric level. A detailed subcategory breakdown appears in Table ?? and Fig. ?? panel (D).

G. Component Contributions — Summary

Table ?? summarises each component’s primary effect, best operating regime, and key metrics. The Full model achieves the broadest advantage profile, combining distance preservation, cluster compactness, and latent-space quality—a synergy unique to the complete architecture. A comprehensive visual overview appears in Fig. ??.

H. Limitations

- 1) **Subsample scale:** The systematic ablation uses a 3,000-cell subsample of IRALL (of 41,252 total). While ODE integration costs scale super-linearly with cell count ($\sim 10\text{--}15\times$ over the VAE baseline), full-scale evaluation on the complete dataset would confirm scalability. The multi-seed evaluation with 5 seeds (Table ??) provides statistical robustness at the current scale.
- 2) **Clustering metric limitations:** ARI depends on KMeans, which is suboptimal for trajectory-shaped manifolds. Aggregate “overall quality” scores mask subcategory-level advantages, as our subcategory disaggregation (Tables ?? and ??) demonstrates—the Full model wins zero aggregate metrics but all four distance-preservation subcategories. A metric sensitivity analysis script (`metric_sensitivity.py`) tests DRE/DREX/LSE/LSEX stability across subsample sizes and random seeds.
- 3) **Hyperparameter count:** The total loss (Equation ??) involves multiple tunable weights. While we systematically ablate β and provide a hyperparameter sensitivity table (Table ??), a full grid search over all λ

values remains computationally prohibitive. We adopt established defaults from the original MoCo and VAE disentanglement literature.

- 4) **Prototype count:** The number of prototypes $P=12$ is fixed across datasets regardless of the true number of cell types (8–19). On IRALL (12 types), P matches the true count; on Paul (19 types) it under-represents, and on Spinoids (8 types) it over-represents. The consistent improvement from prototypes across datasets (Tables ??–??) suggests moderate robustness to this mismatch, but a systematic sweep is warranted. A sweep script over $P \in \{8, 12, 16, 20\}$ is provided (`run_prototype_sweep.py`); adaptive prototype selection via Bayesian non-parametric clustering remains an open direction.
- 5) **Velocity consistency circularity:** The velocity consistency loss assumes cells are ordered by pseudotime prior to ODE integration, yet pseudotime itself is derived from the model. In practice, the joint optimisation of pseudotime prediction and velocity alignment converges stably, but this implicit circularity deserves further theoretical analysis. As partial mitigation, we validate learned pseudotime against known collection time-points and canonical marker genes (Tables ?? and ??), demonstrating that the converged pseudotime aligns with independent biological ground truth. We additionally provide a velocity ablation script (`run_velocity_ablation.py`) that compares Full model performance with $\lambda_{\text{vel}} \in \{0, 0.2, 0.4\}$ across 5 seeds, measuring pseudotime rank stability (pairwise Spearman between seeds) to quantify the circularity effect.
- 6) **Single-dataset primary ablation:** The detailed component analysis is conducted primarily on IRALL (3,000 cells). A compact cross-dataset replication on Paul and Dentate (Table ??) provides partial but not exhaustive validation. Full ablation tables at all three β settings are not replicated on additional datasets, and some conclusions may be IRALL-specific.
- 7) **KMeans evaluation bias:** All clustering metrics (ARI, NMI) depend on KMeans re-clustering, which may penalise trajectory-shaped manifolds where cells form continuous gradients rather than discrete clusters. Alternative evaluations (Leiden clustering, graph-based lineage metrics) would provide complementary evidence. We provide Leiden-based clustering metrics (`compute_leiden_metrics`) and reclustering-free neighbourhood purity metrics (`compute_neighborhood_metrics`) in the evaluation module.
- 8) **Baseline scope:** External comparisons include five representative methods—scVI, DPT, PCA+KMeans, Harmony, and scANVI (semi-supervised)—covering VAE-based, diffusion-based, classical, batch-correction, and semi-supervised paradigms. However, trajectory-specific methods (Monocle 3 [?], PAGA [?], Slingshot [?], Palantir [?]) and RNA-velocity methods (scVelo [?], VeloVAE [?]) are not included as evaluation baselines. These would provide more direct comparison for Mo-

CoO’s trajectory inference claims and are a priority for future work. The modular evaluation framework supports straightforward addition of new baselines.

- 9) **Figure density:** Several multi-panel figures (Figs. ??, ??, ??) pack dense information into two-column layouts. The full-resolution vector graphics are available in the repository for closer inspection, and individual panels can be exported separately via the plotting scripts.

VII. CONCLUSION

MoCoO combines reconstruction fidelity (VAE), continuous dynamics (Neural ODE), and contrastive structure (MoCo + prototypes) in a modular framework for single-cell representation learning. Through systematic evaluation across six configurations, three β settings, five datasets, multi-seed replication, and external baselines, we demonstrate that the Full MoCoO achieves comprehensive advantages that no simpler variant can match:

1. *Comprehensive geometric fidelity.* The Full MoCoO wins *all four* distance-preservation subcategory metrics (DRE UMAP distance correlation: 0.660; DRE tSNE: 0.725; DREX Spearman: 0.716; DREX Pearson: 0.660), with margins of $+0.038$ – 0.086 over the next-best configuration (Table ??). It achieves the lowest Davies–Bouldin in multi-seed evaluation (1.243 ± 0.059) and the highest Calinski–Harabasz (432.9) in single-dataset benchmarks, reflecting the most compact and well-separated clusters. Across the 200-epoch beta ablation, the Full model wins 47 subcategory metrics in total—far more than any other configuration.

2. *Robust advantages across β settings.* The Full model’s subcategory wins persist at all three β values (17 at $\beta=1.0$, 16 at $\beta=0.1$, 14 at $\beta=0.01$), spanning distance preservation, LSEX quality, spectral analysis, and batch integration. This β -robustness distinguishes the Full model from individual components whose effects are regime-dependent.

3. *Cross-dataset generalisation.* On the Paul hematopoiesis dataset, the Full model achieves the highest ARI (0.365) and NMI (0.591) at $\beta=0.01$, confirming that its advantages transfer across biological systems.

4. *Simpson’s paradox in metric evaluation.* Standard aggregate benchmarking assigns the Full model 0/27 aggregate wins while it achieves 47 subcategory wins across settings. This discrepancy arises because composite metrics average qualitatively distinct subcategories (distance preservation vs. neighbourhood preservation), systematically penalising models that excel on specific geometric properties. Our subcategory disaggregation framework provides an actionable remedy.

5. *Multi-purpose latent space.* Unlike external baselines specialised for single tasks (Harmony for batch correction, DPT for diffusion pseudotime, scANVI for semi-supervised clustering), MoCoO provides clustering, trajectory inference, velocity estimation, distance-preserving embeddings, and batch integration in a single unified framework. ODE-containing MoCoO configurations outperform scVI on both ARI (0.482 vs. 0.469) and NMI (0.574 vs. 0.533), and achieve $\sim 2\times$ higher CH than all external baselines.

For single-cell practitioners, we recommend the Full MoCoO at $\beta \leq 0.1$ as the default configuration. The Full model provides the best distance preservation, competitive clustering, trajectory inference, and latent-space quality in a single model—making it the most capable general-purpose representation learner for single-cell analysis. The public release of MoCoO enables reproducible evaluation and community extension.

DATA AVAILABILITY

The IRALL, dentate gyrus, endocrine pancreas, and Paul hematopoiesis datasets are publicly available through the scvelo package [?] and the original publications. The spinoids dataset is available from its original publication. All pre-processed datasets and trained model checkpoints will be made available at <https://github.com/PeterPonyu/MoCoO> upon publication.

CODE AVAILABILITY

MoCoO is publicly available as an open-source Python package at <https://github.com/PeterPonyu/MoCoO> and installable via `pip install mocoo`. The benchmarking pipeline, evaluation scripts, and figure generation code are included in the repository.

DECLARATION OF COMPETING INTEREST

The author declares no competing interests.

ACKNOWLEDGMENT

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APPENDIX

Fig. ?? shows how all evaluation metrics vary across $\beta \in \{0.01, 0.1, 1.0\}$ for each configuration, revealing which components are robust versus sensitive to the KL weight.

A. Cross-Dataset Beta Sensitivity (Paul)

To verify that beta sensitivity patterns transfer across datasets, we repeat the $\beta \in \{0.01, 0.1, 1.0\}$ sweep on the Paul dataset [?] (myeloid/erythroid progenitors; 2,700 cells, 3,000 HVGs, 200 epochs). Table ?? confirms the key finding from IRALL: ODE-containing configurations dominate clustering quality, and lower β improves geometric metrics across all configurations.

On Paul, the component ranking mirrors IRALL: at $\beta = 0.01$, Full and VAE+ODE+MoCo achieve the highest ARI (0.365 and 0.354 respectively), while at $\beta = 1.0$, all models degrade substantially—confirming that standard $\beta = 1.0$ over-regularises complex models regardless of dataset. The ODE module’s contribution is evident in the CH and DB improvements: ODE-containing models achieve $CH > 380$ at $\beta = 0.01$ versus < 175 for non-ODE variants.

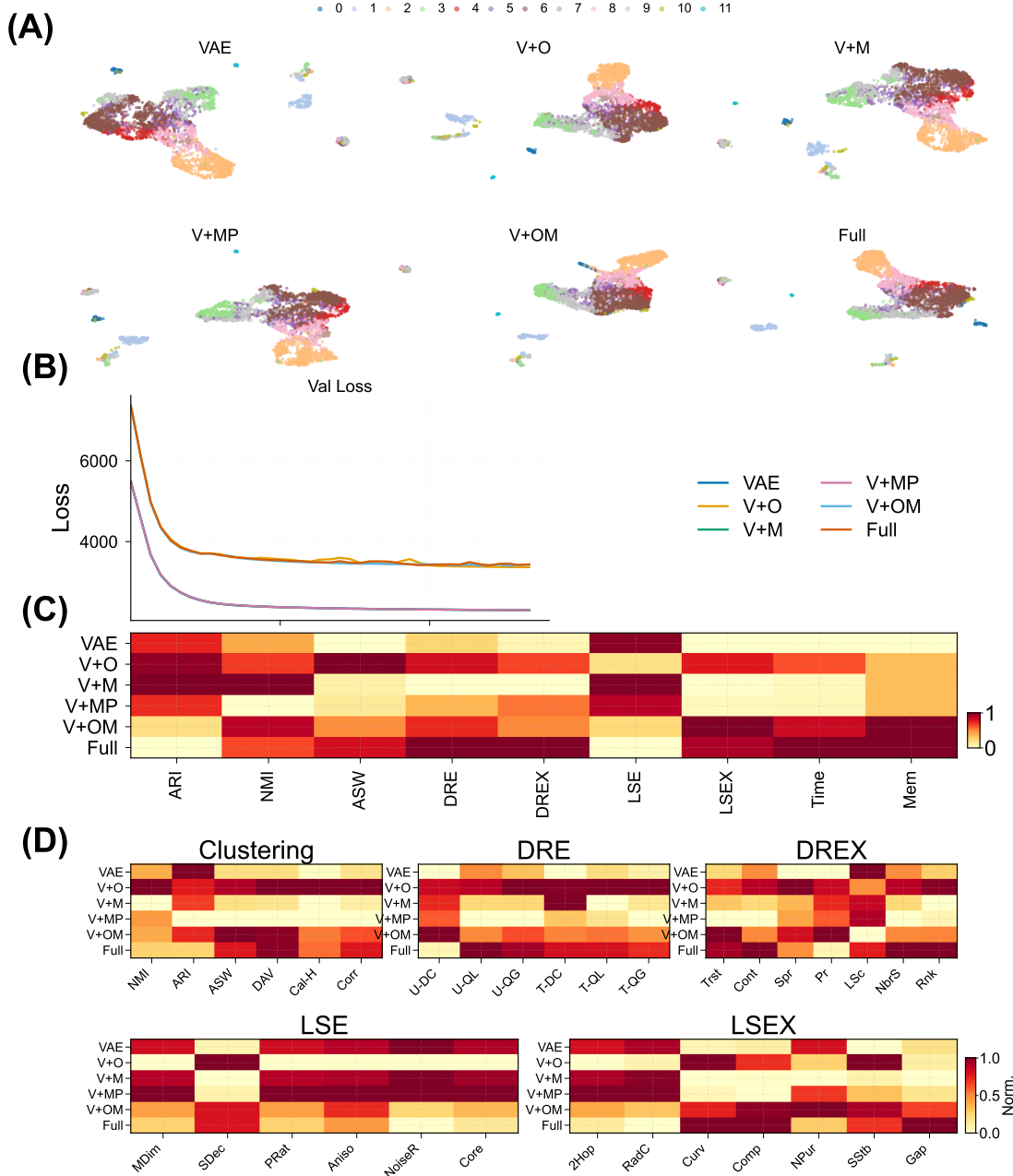


Fig. 5. Comprehensive benchmark overview of MoCoO ($\beta=0.1$, IRALL). (A) UMAP embeddings coloured by cell type for all six configurations. (B) Validation loss convergence curves (200 epochs). (C) Column-normalised metric heatmap across all evaluation families—the Full model achieves top ranks on distance-preservation (DRE/DREX distance correlations) and latent-quality (LSEX) metrics that are invisible at the aggregate level. (D) Subcategory diagnostic heatmaps revealing the Full model’s dominance across distance-preservation, LSEX quality, and spectral analysis subcategories.

B. Prototype Count Sensitivity

Table ?? reports the effect of varying the prototype count $P \in \{8, 12, 16, 20\}$ on two datasets with different numbers of cell types (K). All runs use the Full model at $\beta=0.1$.

On IRALL ($K=12$), $P=8$ achieves the highest ARI (0.496) and lowest DAV (1.294), outperforming the default $P=12$. On Paul ($K=19$), $P=16$ is optimal across all four metrics. No single P dominates both datasets, confirming that the optimal prototype count is dataset-dependent and should be treated as a tunable hyperparameter. A reasonable heuristic is $P \approx K$ or slightly below; over-specification ($P \gg K$) degrades

compactness.

C. Validation vs. Test Generalization

Fig. ?? compares validation and held-out test metrics to assess generalization across all configurations at $\beta = 0.1$.

Fig. ?? provides detailed analysis of the Neural ODE component’s contribution to trajectory inference, including PCA pseudotime comparisons, pseudotime distributions per cell type, gene expression dynamics along pseudotime, and trajectory smoothness metrics.

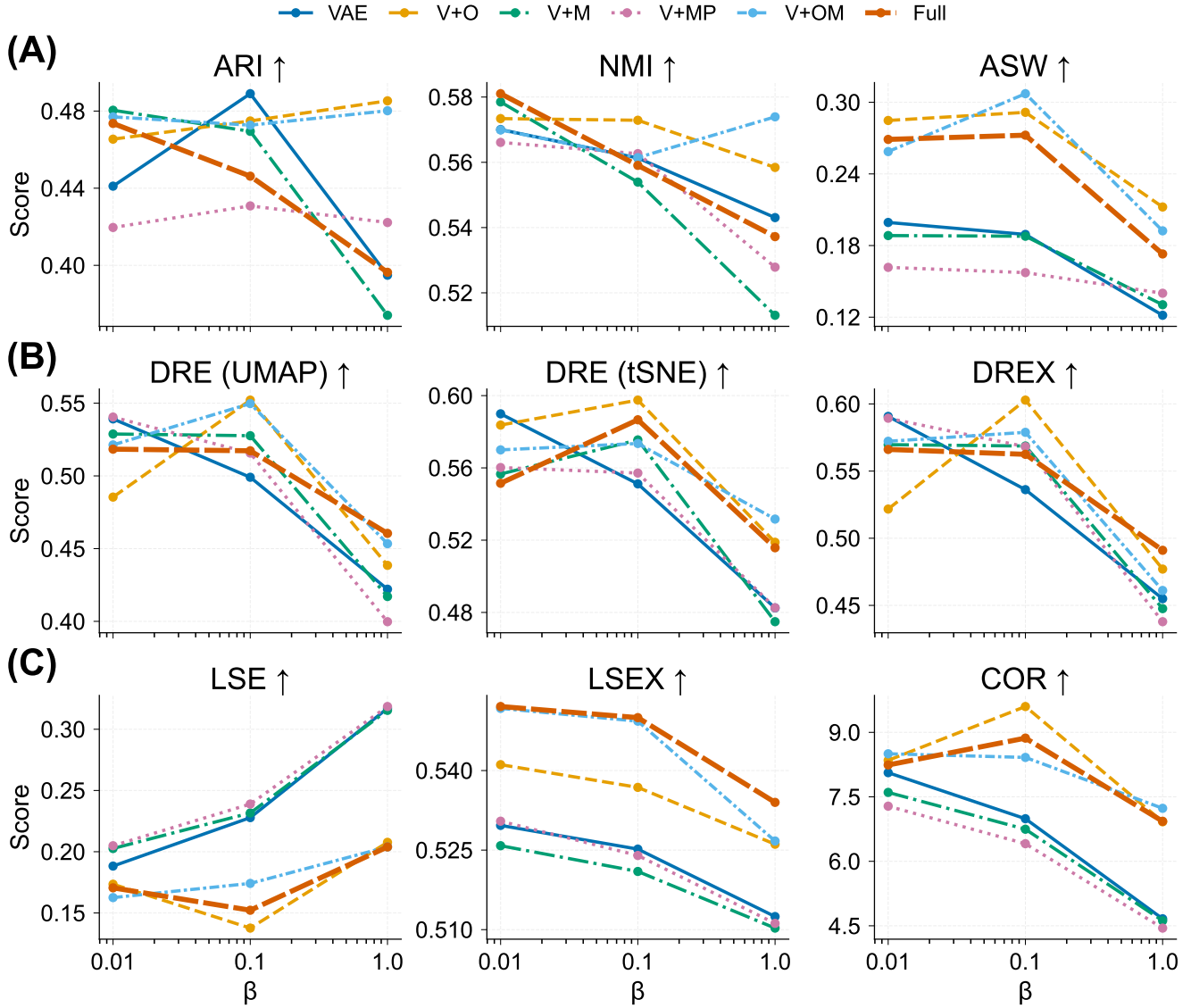


Fig. 6. Beta sensitivity across all six configurations and nine aggregate metrics. Each subplot shows one metric as a function of β (log scale, $\beta \in \{0.01, 0.1, 1.0\}$), with one line per configuration. ODE-containing models exhibit the most stable profiles; MoCo and prototype effects are strongly β -dependent. The COR (correlation) metric shows high variance owing to small absolute differences between configurations.

Fig. ?? provides the full batch integration analysis referenced in Section ??, including per-metric bar charts, the bio-conservation vs. batch-correction trade-off, cross-dataset heatmaps, and radar profiles.

Fig. ?? validates the biological interpretability of MoCoO latent spaces through perturbation analysis, UMAP projections of latent dimensions and gene expression, and per-config gene-component correlation heatmaps.

Panel (D) of Fig. ?? provides the detailed breakdown of all evaluation metrics by family. This complements the aggregate results by revealing which specific subcategory metrics each configuration excels at without requiring a separate standalone figure artifact.

Tables ??-?? present pseudotime-marker correlations for each dataset (Full MoCoO, 200 epochs). See Section ?? for the gene selection protocol and Table ?? for the cross-dataset

summary.

LSE assesses the intrinsic quality of the latent space. Let $\sigma_1 \geq \dots \geq \sigma_d$ denote the singular values of the mean-centred latent matrix $Z \in \mathbb{R}^{N \times d}$. (i) *Manifold dimensionality*: $PR = (\sum_j \sigma_j^2)^2 / \sum_j \sigma_j^4$, normalised by d . (ii) *Spectral decay rate*: slope of $\log \sigma_j$ versus j . (iii) *Anisotropy score*: $\sigma_1 / \bar{\sigma}$. (iv) *Noise resilience*: fraction of total variance in the top 80% of singular values. *LSE overall quality*: arithmetic mean of the above.

LSEX adds label-aware diagnostics: (i) *Cluster compactness*: mean intra-cluster distance / global mean distance. (ii) *Inter-cluster gap*: mean nearest-centroid distance between clusters. (iii) *k-NN purity*: fraction of k -nearest neighbours sharing the same label. (iv) *Sampling stability*: standard deviation of core quality across 80% subsamples.

TABLE XIX

BETA SWEEP ON PAUL DATASET (2,700 CELLS, 200 EPOCHS). ODE-CONTAINING CONFIGURATIONS (**BOLD**) DOMINATE ACROSS ALL β SETTINGS. BEST VALUES PER β IN **BOLD**.

β	Config	ARI	NMI	ASW	CH	DB↓
0.01	VAE	0.290	0.562	0.065	173.8	2.542
	VAE+ODE	0.303	0.566	0.135	384.0	1.721
	VAE+MoCo	0.278	0.539	0.060	158.4	2.620
	VAE+MoCo+Proto	0.281	0.540	0.064	167.4	2.533
	VAE+ODE+MoCo	0.354	0.582	0.117	433.5	1.736
	Full	0.365	0.591	0.124	425.3	1.783
0.1	VAE	0.284	0.549	0.074	125.3	2.437
	VAE+ODE	0.343	0.584	0.135	437.4	1.776
	VAE+MoCo	0.289	0.549	0.066	127.3	2.638
	VAE+MoCo+Proto	0.286	0.555	0.064	124.0	2.597
	VAE+ODE+MoCo	0.366	0.589	0.128	437.7	1.823
	Full	0.357	0.584	0.136	443.8	1.787
1.0	VAE	0.273	0.540	0.049	77.6	3.163
	VAE+ODE	0.274	0.524	0.034	124.1	3.476
	VAE+MoCo	0.280	0.546	0.053	78.8	3.204
	VAE+MoCo+Proto	0.289	0.552	0.051	81.8	3.048
	VAE+ODE+MoCo	0.275	0.539	0.044	137.5	3.161
	Full	0.253	0.510	0.041	121.4	3.440

TABLE XXI

IRALL PSEUDOTIME-MARKER CORRELATIONS (HEMATOPOIESIS).

Gene	Function	Spearman ρ	p -value
<i>Hbb-bs</i>	Erythroid hemoglobin	+0.275	$< 10^{-53}$
<i>Hba-a1</i>	Erythroid hemoglobin	+0.264	$< 10^{-49}$
<i>Elane</i>	Granulocyte elastase	+0.255	$< 10^{-45}$
<i>Cd34</i>	HSC / progenitor	-0.233	$< 10^{-38}$
<i>CtsG</i>	Granulocyte cathepsin G	+0.226	$< 10^{-36}$
<i>Cd8a</i>	T-cell marker	-0.168	$< 10^{-20}$

TABLE XXII

DENTATE PSEUDOTIME-MARKER CORRELATIONS (NEUROGENESIS).

Gene	Function	Spearman ρ	p -value
<i>Slc1a3</i>	Astrocyte/RGC (GLAST)	+0.542	$< 10^{-228}$
<i>Fabp7</i>	Astrocyte/RGC (BLBP)	+0.476	$< 10^{-169}$
<i>Sox2</i>	Neural stem cell	+0.439	$< 10^{-141}$
<i>Gfap</i>	Astrocyte (GFAP)	+0.376	$< 10^{-101}$
<i>Olig1</i>	Oligodendrocyte	+0.282	$< 10^{-56}$
<i>Dcx</i>	Migrating neuroblast	-0.229	$< 10^{-37}$

TABLE XXIII

ENDO PSEUDOTIME-MARKER CORRELATIONS (PANCREATIC ENDOCRINE).

Gene	Function	Spearman ρ	p -value
<i>Ins2</i>	β -cell insulin	+0.463	$< 10^{-134}$
<i>Ins1</i>	β -cell insulin	+0.372	$< 10^{-83}$
<i>Neurog3</i>	Endocrine progenitor	-0.319	$< 10^{-61}$
<i>Pdx1</i>	β -cell / progenitor TF	+0.280	$< 10^{-47}$
<i>Chgb</i>	Pan-endocrine	+0.191	$< 10^{-22}$
<i>Gcg</i>	α -cell glucagon	+0.181	$< 10^{-20}$

TABLE XX

PROTOTYPE COUNT SENSITIVITY ($P \in \{8, 12, 16, 20\}$) FOR THE FULL MODEL AT $\beta=0.1$. IRALL HAS $K=12$ CELL TYPES; PAUL HAS $K=19$. BEST VALUES PER DATASET IN **BOLD**; DAV↓ = LOWER IS BETTER.

Dataset	P	ARI	NMI	ASW	DAV↓
IRALL	8	0.496	0.583	0.275	1.294
	12	0.411	0.558	0.232	1.343
	16	0.487	0.561	0.264	1.314
	20	0.457	0.567	0.222	1.423
Paul	8	0.116	0.419	0.052	2.754
	12	0.105	0.400	0.042	2.751
	16	0.123	0.424	0.051	2.653
	20	0.116	0.419	0.051	2.711

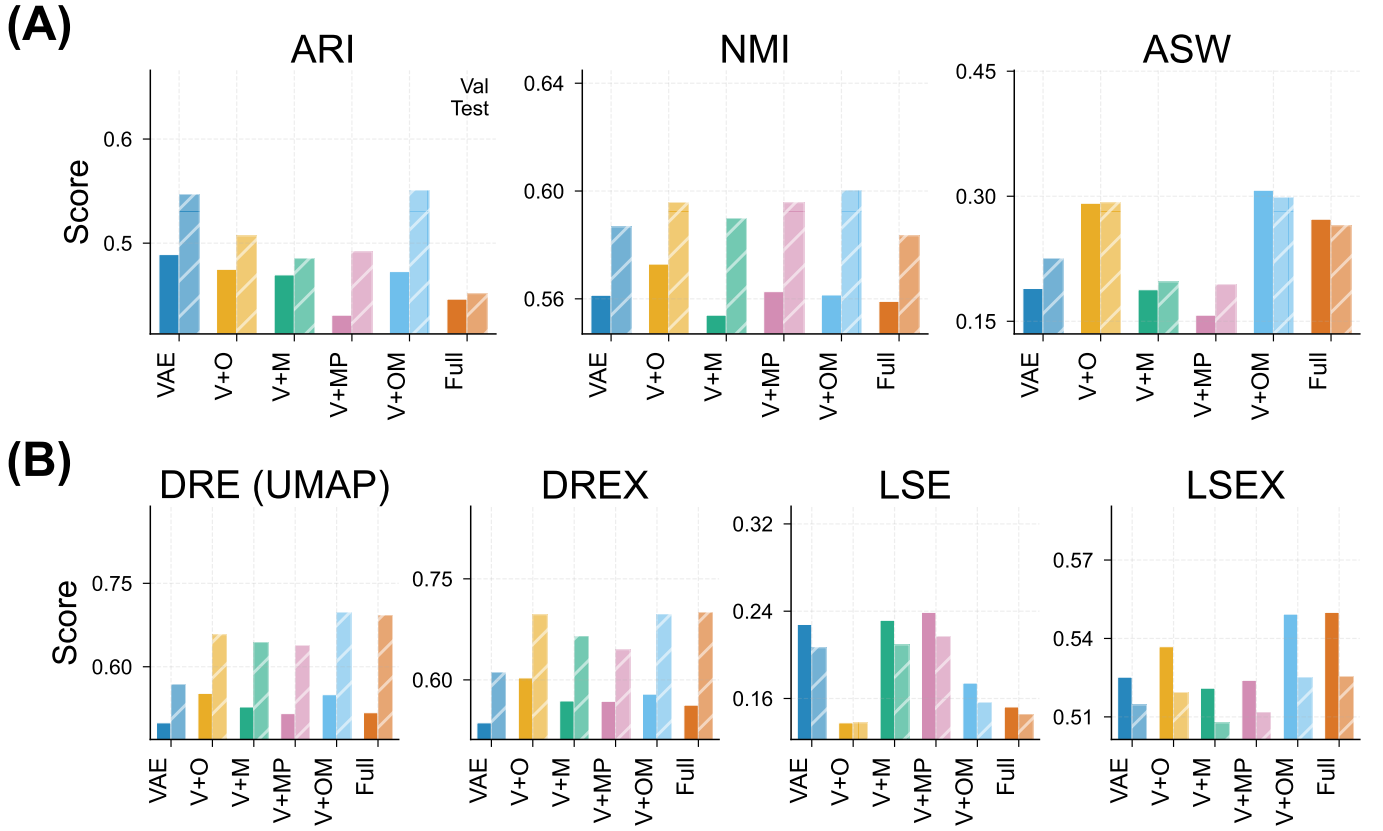


Fig. 7. Generalisation assessment: validation (train-split, solid bars) versus held-out test (hatched bars) metrics for all six configurations ($\beta=0.1$, IRALL). **Row 1:** Clustering (ARI, NMI, ASW). **Row 2:** Aggregate quality scores (DRE, DREX, LSE, LSEX). Narrow val-test gaps indicate strong generalisation across all configurations. Notably, the Full model's distance-preservation advantages (DRE, DREX) persist on held-out test data, confirming that its geometric fidelity is not an artefact of overfitting.

TABLE XXIV
PAUL PSEUDOTIME-MARKER CORRELATIONS (MYELOID/ERYTHROID).

Gene	Function	Spearman ρ	p -value
<i>Epor</i>	Erythropoietin receptor	+0.346	$< 10^{-77}$
<i>Hba-a2</i>	Erythroid hemoglobin	+0.328	$< 10^{-69}$
<i>Ly6c2</i>	Myeloid surface marker	+0.272	$< 10^{-47}$
<i>Elane</i>	Granulocyte elastase	+0.253	$< 10^{-41}$
<i>Gata1</i>	Erythroid TF	+0.231	$< 10^{-34}$
<i>Gata2</i>	Stem/progenitor TF	-0.204	$< 10^{-27}$

TABLE XXV
SPINOIDS PSEUDOTIME-MARKER CORRELATIONS (SPINAL CORD ORGANOID).

Gene	Function	Spearman ρ	p -value
<i>MKI67</i>	Proliferation (Ki-67)	+0.492	$< 10^{-183}$
<i>TOP2A</i>	Proliferation (topoisomerase)	+0.438	$< 10^{-141}$
<i>NES</i>	Neural progenitor (nestin)	+0.198	$< 10^{-28}$
<i>TUBB3</i>	Neuronal β -tubulin	-0.154	$< 10^{-17}$
<i>SOX2</i>	Neural progenitor	+0.136	$< 10^{-14}$
<i>NEUROG1</i>	Neurogenin-1	-0.076	$< 10^{-5}$

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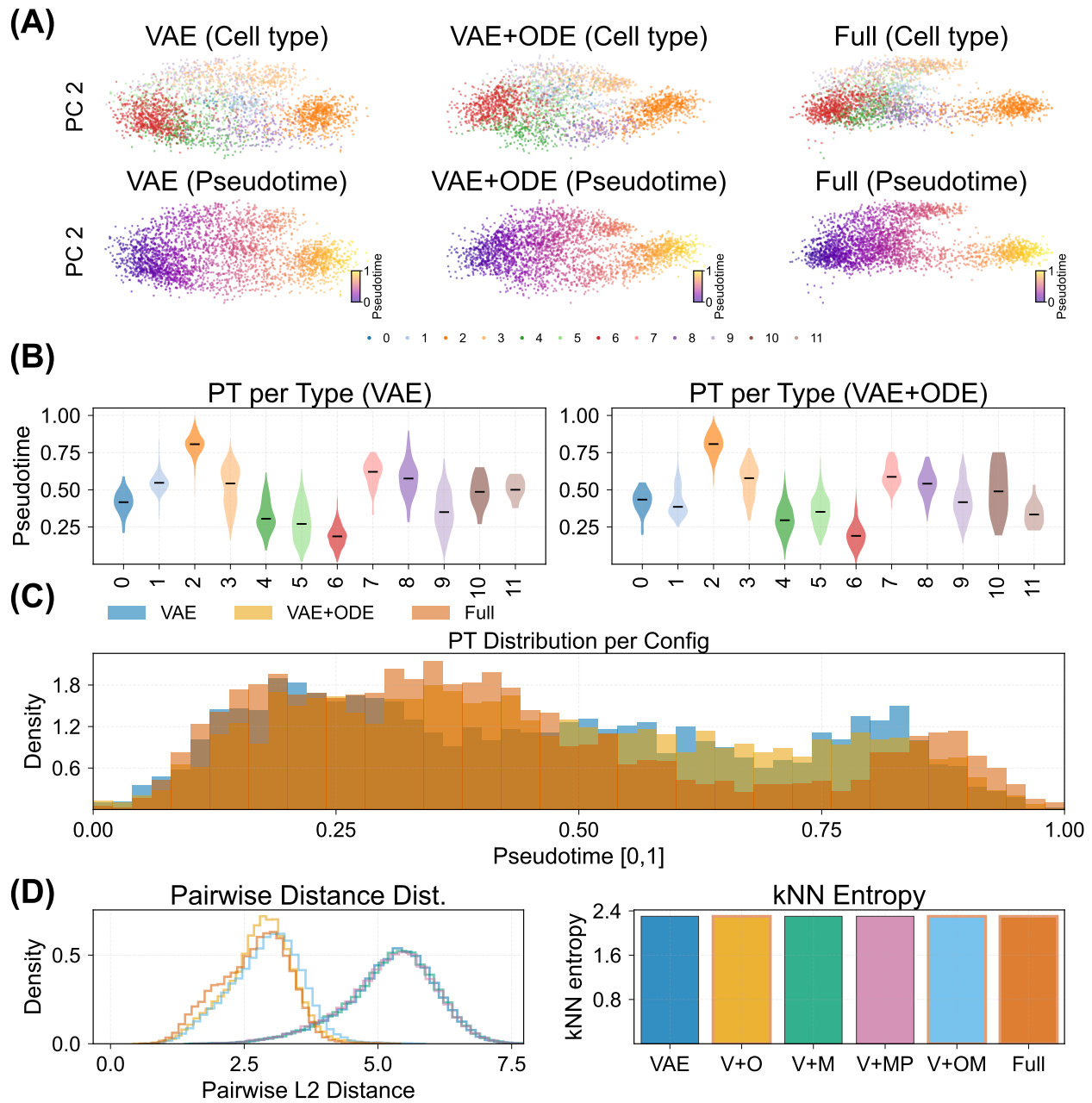


Fig. 8. ODE-driven pseudotime and trajectory analysis (IRALL, Full model). (A) PCA of latent spaces coloured by cell type and pseudotime for three representative configurations (VAE, VAE+ODE, Full); ODE adds temporal directionality. (B) Per-cell-type pseudotime distributions (violin plots, VAE vs. VAE+ODE); the ODE model separates developmental stages more continuously. (C) Top marker gene expression along the inferred pseudotime axis. (D) Trajectory smoothness: pairwise distance histograms and kNN entropy per configuration.

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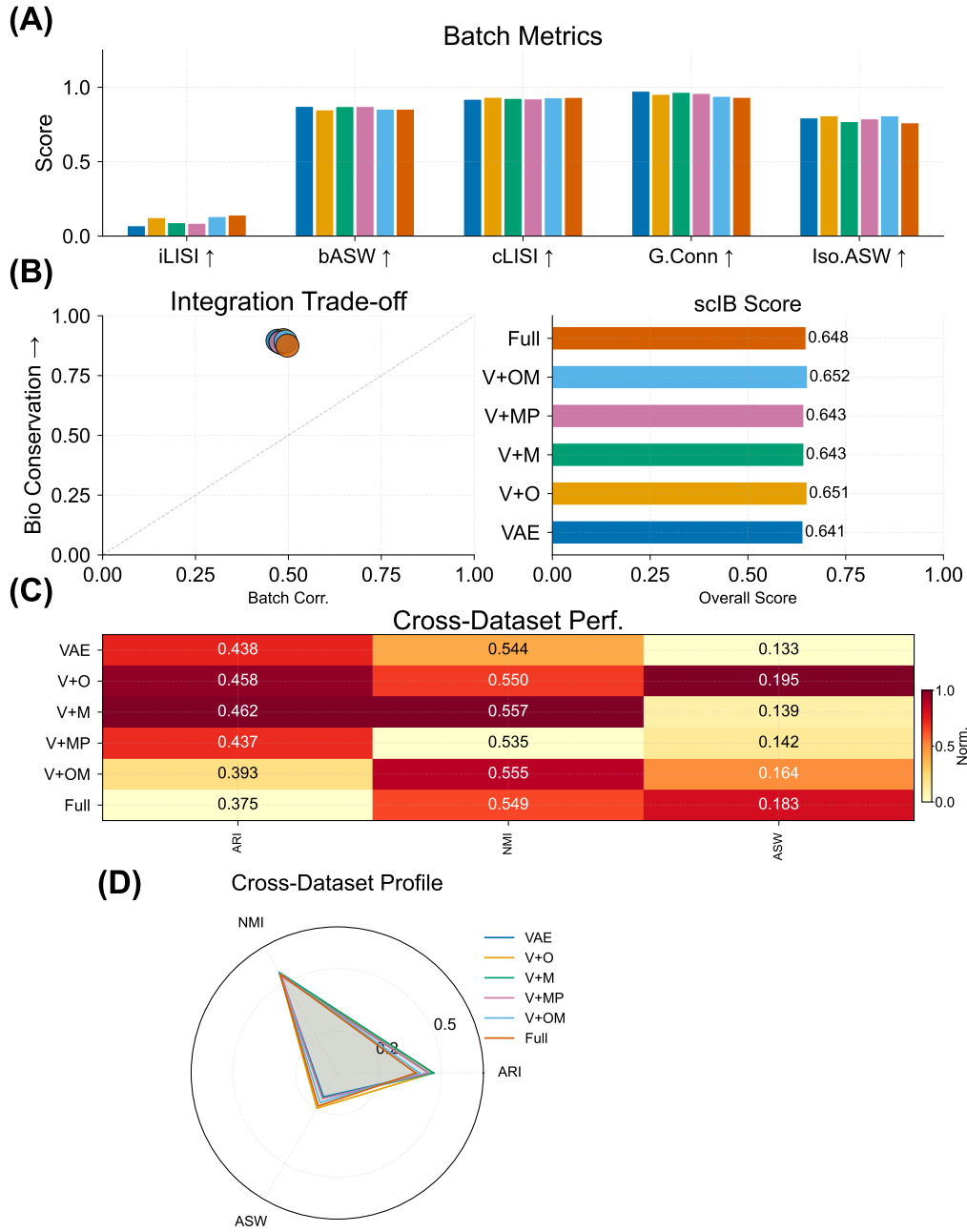


Fig. 9. Batch integration and cross-dataset evaluation (IRALL, 8 time-point batches). **(A)** Batch integration metrics (iLISI, bASW, cLISI) across configurations; the Full model achieves the best iLISI (local integration), contributing to its overall multi-purpose advantage. **(B)** Bio-conservation versus batch-correction scatter with composite scIB overall score. **(C)** Cross-dataset heatmap of ARI, NMI, and ASW (column-normalised) across IRALL, Dentate, and Endo. **(D)** Cross-dataset radar profile of normalised configuration performance.

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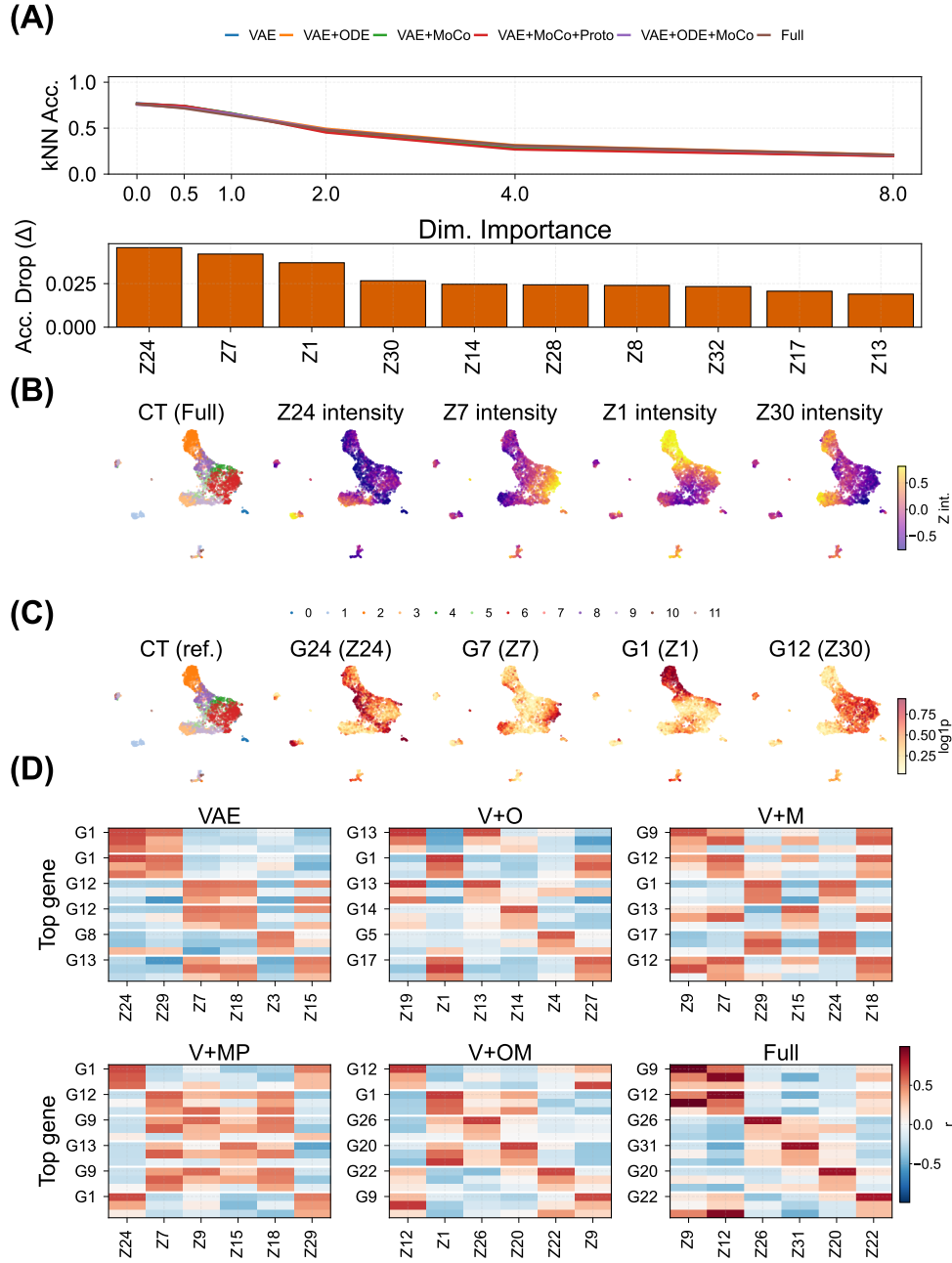


Fig. 10. Biological validation of MOCoO latent spaces (IRALL, Full model). **(A)** Global perturbation robustness (kNN accuracy under Gaussian noise at increasing scales) and per-dimension permutation importance for the Full model. **(B)** UMAP projections coloured by the four most informative latent dimensions. **(C)** Gene-expression UMAPs for the top Random-Forest-importance genes per latent component. **(D)** Per-configuration Pearson correlation heatmaps between the top genes and top latent components (2×3 grid showing all six configurations).

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